



IMPACT ASSESSMENT REPORT

Economic Trasformation Initiative - Gilgit-Baltistan (ETI-GB)

Date: 30th January 2026

Disclaimer

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We remain committed to supporting evidence-based learning and look forward to continued collaboration with ETI-GB in advancing rural transformation in Gilgit-Baltistan.

With warm regards,



Himatullah

Chief Executive Officer

HIMAT Consulting Pvt Ltd

Summary Table

Table A: COI Impact Scorecard (ETI-GB)

Indicator Code / Name	Indicator Definition (Log-frame / COI)	Baseline	Log-frame Target	Reference Value Treatment Group (2025)	Reference Value Control Group (2025)	Net Effect / Attribution	Assessment Status
CI 1.1.1 – Secure access to land	Average irrigated land per HH (acres)	0.60 acres/HH	Expansion of irrigated landholding	1.5 acres/HH	0.60 acres/HH	+1.3 acres attributable to ETI	Exceeded
CI 1.2.1 – Improved access to land (%)	HHs reporting improved land access	0	50,000 HHs expand access	89.20	70.20	+19.0 pp due to ETI-GB	Exceeded
CI 1.1.3 – VACs accessing improved inputs (%)	HHs accessing inputs via VACs	0	Increased institutional access	69.50	Not observed	Institutional effect of VACs	Achieved
CI 1.2.2 – Adoption of improved practices (%)	HHs adopting improved agri practices	0	≥50% HHs adopt	78.50	Negligible	Extension + VC driven adoption	Exceeded
CI 1.2.4 – Increased agri production (%)	HHs reporting increased production	0	≥25% increase	60.90	27.00	+33.9 pp attributable	Exceeded
Logframe Income Indicator – Income incidence (%)	HHs increasing agri/rural income by ≥25	0	≥50% HHs	86.60	73.00	+13.6 pp net impact	Exceeded
Logframe Income Indicator – Mean HH income (PKR/month)	Average monthly HH income	Not measured	Increased income	55,000	50,000	+10% income premium	Achieved
Average agricultural income (PKR/month)	Monthly agri income per HH	3,750*	Increase vs baseline	16,830	8,200	+105% treatment premium	Exceeded
Average agricultural income (PKR/year)	Annual agri income per HH	45,000	≥25% increase	201,960	98,400	+348% over baseline	Exceeded

CI 2.2.6 – Improved market access (%)	HHs reporting reduced travel time / costs	Not measured	Improved access	91.30	N/A	FMR-driven access gains	Achieved
CI 2.2.6 – Time saved per trip (hours)	Avg. reduction in market travel time	–	Reduced transaction costs	3.5 hours	–	Infrastructure effect	Achieved
CI 2.1.3 – Producer organizations supported	Functional VACs / POs established	0	Establish POs	162 VACs	0	Institutional strengthening	Achieved
CI 2.1.4 – HHs affiliated with POs	HH members of VACs / POs	0	Increased membership	40,613 HHs	0	Scale effect of institutions	Achieved
CI 2.2.3 – POs with buyer contracts (%)	Formal FO–buyer linkages	0	220 contracts	33.3% of POs	0	VC partially operational	Partially Achieved
CI 2.2.2 – PO profitability	Avg. annual profit per PO	0	Increased profits	PKR 212,368 / year	N/A	Strong VC outcome	Exceeded
CI 1.2.8 – Women dietary diversity	MDD-W score	Not measured	≥60 women	DDS 9.07 /10	8.95 /10	Positive nutrition effect	Achieved
CI 1.2.9 – Nutrition & hygiene KAP	Food hygiene behavior index	Not measured	Improved KAP	FHBI 2.07	1.89	Behavioral change	Achieved
Welfare – Medical access (%)	HHs able to seek care when needed	Limited	Improved access	64.80	55.10	Welfare spillover	Achieved
Goal-level – Poverty reduction (%)	HHs below poverty line (PSC)	34.60	≥15 reduction	3.61	5.60	Large poverty exit	Exceeded
Resilience – HHs with savings (%)	HHs reporting savings	Low	Improved resilience	30.90	26.00	Income → savings pathway	Achieved

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List of Abbreviations

Abbreviation	Full Form
ETI-GB	Economic Transformation Initiative Gilgit-Baltistan
GB	Gilgit-Baltistan
GGOB	Government of Gilgit-Baltistan
IFAD	International Fund for Agricultural Development
AICS	Italian Agency for Development Cooperation
P&DD	Planning and Development Department
PMU	Programme Management Unit
ILD	Irrigation and Land Development
ToC	Theory of Change
FMR	Farm-to-Market Roads
VAC	Village Agriculture Cooperative
VC	Value Chain
VCD	Value Chain Development
VCF	Value Chain Fund
VCA	Value Chain Assessment / Initiatives
VCSF	Value Chain Support Fund
WUAs	Water User Associations
O&M	Operations and Maintenance
ADPs	Annual Development Plans
SDR	Special Drawing Rights
OECD-DAC	Organization for Economic Co-operation and Development – Development Assistance Committee
TORs	Terms of Reference
PDR	Programme Design Report
AMEL	Activity Monitoring, Evaluation, and Learning
MEAL	Monitoring, Evaluation, Accountability, and Learning
FGD	Focus Group Discussion
KII	Key Informant Interview
HHs	Households
HDDS	Household Dietary Diversity Score
PO	Producer Organization
PCU	Programme Coordination Unit
RCC	Reinforced Cement Concrete
PPPP / 4P	Public–Private–Producer Partnership
GIS	Geographic Information System
DFID	Department for International Development (UK)
PSLM	Pakistan Social and Living Standards Measurement Survey
BISP	Benazir Income Support Programme
SOPs	Standard Operating Procedures
FHBI	Food Hygiene Behavior Index

Executive Summary

This report presents the findings of the Impact Assessment of the Economic Transformation Initiative–Gilgit-Baltistan (ETI-GB), implemented by the Government of Gilgit-Baltistan with co-financing from the International Fund for Agricultural Development (IFAD) and the Italian Agency for Development Cooperation (AICS). The assessment evaluates programme performance against the ETI-GB logframe and the Core Outcome Indicators (COI) using evidence collected in 2025. A mixed-methods, quasi-experimental design was applied, combining treatment–control household survey data with qualitative evidence from Focus Group Discussions, Key Informant Interviews, and field observations. Overall, the assessment finds that ETI-GB has met or exceeded the majority of its logframe and COI targets, particularly those related to access to productive assets, agricultural production, employment, income growth, poverty reduction, market access, and institutional strengthening. The strongest and most consistent results are observed where interventions were delivered in an integrated manner, confirming the programme’s theory of change.

Access to Land

Beneficiaries gaining increased secure access to land (CI 1.1.1) and households reporting improved access to land and water for production (CI 1.2.1) show strong progress under ETI-GB. Average irrigated land per household increased from 0.60 acres at baseline to 1.5 acres in 2025, representing a net gain of 0.9 acres attributable to irrigation and land development interventions. Correspondingly, 89.2 percent of beneficiary households reported improved access to land and water for production, compared to 70.2 percent among control households, exceeding logframe expectations. These improvements established a durable productive asset base in previously water-scarce areas.

Technology Adoption and Agricultural Services

Strong results are observed for households adopting new or improved agricultural practices (CI 1.2.2) and Village Agriculture Cooperatives accessing improved inputs and technology packages (CI 1.1.3). A total of 78.5 percent of beneficiary households adopted improved agricultural technologies and practices, far exceeding the logframe benchmark of 50 percent and reflecting a substantial improvement from negligible adoption at baseline. Access to extension and advisory services also improved, with 27.5 percent of beneficiary households reporting access compared to 7.4 percent of control households. These outcomes demonstrate that ETI-GB effectively addressed key knowledge and input constraints faced by farmers in remote mountain contexts.

Agricultural Production and Productivity

ETI-GB interventions translated improved asset access and technology adoption into significant gains in production. Households reporting an increase in agricultural production (CI 1.2.4) reached 60.9 percent among treatment households, compared to 27.0 percent among control households, yielding a net impact of nearly 34 percentage points. This result substantially exceeds the logframe target of a 25 percent increase in production and confirms strong progress in agricultural productivity driven by expanded irrigation, increased cropping intensity, and improved farm management practices.

Employment Creation and Income Growth

Clear progress is evident for beneficiaries with new jobs or employment opportunities (CI 2.2.1). 86.6 percent of beneficiary households reported engagement in new or additional employment and income-generating activities linked to ETI-GB interventions, including on-farm employment, value-

chain activities, cooperative operations, and rural enterprises. Income outcomes further reinforce these employment effects. Households reporting increased income from agricultural and rural economic activities (Logframe Income Indicator) reached 86.6 percent, and nearly 60 percent of beneficiary households experienced income gains of 50 percent or more, significantly exceeding the logframe target of a minimum 25 percent income increase for at least half of beneficiaries. Mean monthly household income among beneficiaries reached PKR 67,897, compared to PKR 63,039 among control households.

Market Access and Connectivity

The Farm-to-Market Roads component delivered strong results for households reporting improved physical access to markets (CI 2.2.6). Beneficiary households reported an average time saving of 3.5 hours per market trip, and 91.3 percent perceived improved access to markets, storage, and services. These outcomes confirm that improved connectivity significantly reduced transaction costs and enabled more frequent market engagement, amplifying production and income gains.

Institutional Strengthening and Value Chain Development

ETI-GB achieved substantial progress in strengthening rural institutions. Rural producers' organizations supported (CI 2.1.3) reached 162 Village Agriculture Cooperatives, while households affiliated with producer organizations (CI 2.1.4) totaled 40,613 member households, compared to zero at baseline. All assessed producer organizations reported increased profitability, with an average annual profit of PKR 212,368, demonstrating strong achievement for supported rural enterprises reporting increased profits (CI 2.2.2). Progress toward formal market integration under producer organizations with contracts or buyer linkages (CI 2.2.3) reached 33.3 percent, representing meaningful progress from a zero baseline, though indicating scope for further strengthening.

Nutrition, Health, and Household Well-Being

ETI-GB generated positive welfare spillovers. Women reporting minimum dietary diversity (CI 1.2.8) achieved a Women's Dietary Diversity Score of 9.07 among beneficiary households, compared to 8.95 among control households. Improvements were also observed for households with improved nutrition and hygiene-related knowledge, attitudes, and practices (CI 1.2.9), with beneficiary households recording a higher Food Hygiene Behavior Index (2.07 vs. 1.89). Access to healthcare improved alongside income and mobility gains. Households reporting improved ability to access health services showed better outcomes among beneficiaries (64.8 percent) compared to control households (55.1 percent), indicating improved well-being and reduced vulnerability.

Poverty Reduction and Resilience

At goal level, reduction in the proportion of households living in poverty (Goal-level Indicator) substantially exceeded expectations. Using the BISP Poverty Scorecard, poverty incidence among beneficiary households declined from 34.6 percent at baseline to 3.61 percent at the time of this assessment, representing a reduction of more than 30 percentage points, well above the logframe target of a 15 percent reduction. Beneficiary households also performed better than control households, which recorded a poverty headcount of 5.60 percent.

Overall Assessment

Taken together, the evidence confirms that ETI-GB has delivered strong, consistent, and attributable progress across its Outcome Indicators. Most indicators were fully achieved or exceeded, particularly those related to productive assets, production, employment, income growth, poverty reduction,

market access, and institutional strengthening. Areas requiring continued attention include deepening formal buyer linkages for producer organizations. Overall, ETI-GB demonstrates that integrated, asset-based, and market-oriented rural development interventions can generate sustained economic transformation in remote mountain regions, providing a strong empirical basis for scaling and replication.

Chapter One: Introduction and Background

1. Programme Overview

The Economic Transformation Initiative Gilgit-Baltistan (ETI-GB) is a flagship programme of the Government of Gilgit-Baltistan, co-funded by IFAD and the Italian Agency for Development Cooperation (AICS). The programme seeks to improve rural incomes and reduce poverty and malnutrition across Gilgit-Baltistan. With a total investment of USD 133.72 million (IFAD USD 67.00 million, AICS USD 21.12 million, GoGB USD 23.63 million, and beneficiary contribution USD 21.97 million), ETI-GB was originally approved for 2015-2022 at USD 120.15 million and later extended to 2029 with the Executive Committee of the National Economic Council (ECNEC)'s approval and concurrence of co-financiers. The programme aims to benefit 100,000 rural households through three mutually reinforcing components: economic infrastructure development, support services for value chain development, program management, and policy support.

2. Rationale and Objectives of the Impact Assessment

To systematically assess the outcomes and emerging impacts of ETI-GB, an impact assessment study was conducted in 2025. The primary purpose of the assessment was to examine changes in the socio-economic conditions and living standards of beneficiary households and to assess the extent to which these changes can be attributed to ETI-GB interventions.

The assessment also aimed to capture beneficiaries' perceptions of project-induced change and to generate lessons that can inform future programme design, scaling, and policy engagement in Gilgit-Baltistan and similar mountain contexts. Data collection for the study was carried out between October and November 2025, and the findings presented in this report are based on a detailed analysis of quantitative and qualitative data collected during this period.

3. Programme Components and Intervention Design

ETI-GB adopts an integrated approach to rural economic transformation by combining investments in productive infrastructure with improved market access and institutional strengthening. The programme's core intervention areas include Irrigation and Land Development (ILD), Farm-to-Market Roads (FMRs), and rural bridges, which together aim to enhance agricultural productivity, connectivity, and livelihoods across Gilgit-Baltistan. Under the Irrigation and Land Development (ILD) component, 88 schemes have been completed, resulting in the development of 42,238 acres of command area and directly benefiting 30,506 households. These investments have significantly expanded irrigated land, improved water availability, and strengthened the productive base of rural households. To improve physical connectivity and access to markets and services, ETI-GB has completed 92 Farm-to-Market Road (FMR) schemes, covering a total length of 434.7 kilometers. These road investments have enhanced mobility and market integration for an estimated 15,476 households, reducing transport costs and post-harvest losses while improving access to inputs, markets, and essential services. In addition, the programme has completed 13 rural bridge schemes, with a cumulative bridge length of 388.15 meters. These bridges have played a critical role in improving year-round connectivity, particularly in remote and geographically constrained areas, facilitating safer movement of people and goods and strengthening linkages between villages and markets.

Complementing physical infrastructure, 162 value chain development initiatives have been implemented across Gilgit-Baltistan, reaching 40,613 households. These interventions support

producer organizations, improve access to quality inputs, promote nursery and orchard development, and strengthen market linkages, with particular attention to women and disadvantaged groups.

4. Integrated Approach and Expected Contribution to Rural Transformation

Taken together, ETI-GB's integrated design reflects a comprehensive approach to rural transformation in a challenging mountain context. By simultaneously addressing constraints related to water availability, land productivity, physical connectivity, and market access, the programme seeks to generate sustained improvements in agricultural productivity, household incomes, employment opportunities, and social inclusion.

The impact assessment presented in this report examines the extent to which these objectives are being realized and explores the pathways through which ETI-GB interventions contribute to socio-economic change in Gilgit-Baltistan.

Chapter Two: Methodology

1. Design of Impact Assessment

The impact assessment applies a quasi-experimental design to assess the extent to which observed changes in household welfare, agricultural performance, and market engagement can be attributed to the ETI-GB. Given the nature of the programme—comprising large-scale, area-based infrastructure interventions such as irrigation and land development, farm-to-market roads, and associated value chain support—random assignment was neither feasible nor appropriate. The assessment therefore relies on a treatment–control comparison framework to establish a credible counterfactual.

Treatment households were selected from villages where ETI-supported interventions had been completed and were operational at the time of the study. Control households were drawn from villages with comparable agroecological conditions, livelihood systems, and market access constraints, but without exposure to ETI interventions. The conceptual logic of this design, including the relationship between ETI inputs, intermediate outcomes, and longer-term impacts, is illustrated in the impact assessment conceptual framework presented in Figure 1.

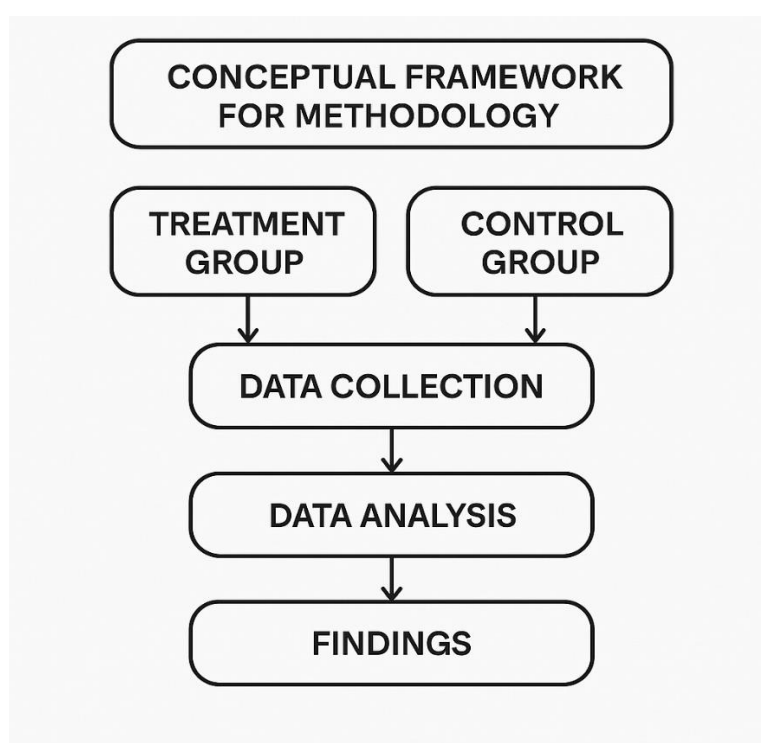


Figure 1: Conceptual Framework

The use of control groups provides a reference point for distinguishing programme effects from broader trends affecting Gilgit-Baltistan, including climatic variability, migration patterns, price fluctuations, and parallel investments unrelated to ETI. Without a valid comparison group, changes in income, productivity, food security, or mobility could not be confidently attributed to ETI support. By combining treatment–control comparisons with triangulation from qualitative evidence, the assessment strengthens causal inference while remaining consistent with the operational realities of infrastructure-led rural development.

2. Sampling Framework

The sampling framework is designed to ensure adequate representation of ETI-GB interventions while enabling statistically and analytically valid comparisons between beneficiary and non-beneficiary populations. The approach follows the parameters defined in the inception report and reflects the geographical spread, intervention diversity, and implementation status of ETI-supported activities at the time of data collection.

District selection focuses on areas where ETI-supported irrigation and land development schemes, farm-to-market roads, and value chain interventions are operational or functionally active. The selected districts capture variation in agro ecological conditions, livelihood systems, and market access contexts across Gilgit-Baltistan. Within each selected district, treatment villages are identified using official programme records confirming that interventions are completed and in use. Control villages are selected from neighboring or comparable areas within the same districts that share similar agro ecological characteristics, production systems, and market access conditions but have not received ETI support. This approach ensures that observed differences between treatment and control groups can reasonably be attributed to programme interventions rather than underlying contextual variation.

Household sampling within villages is conducted using beneficiary registers for treatment communities and complete household listings for control communities. Treatment households are proportionately sampled to reflect actual exposure to irrigation and land development, value chain support, and farm-to-market road influence zones, including households benefiting from single interventions as well as those exposed to multiple ETI components. Control households are randomly selected to mirror, as closely as possible, the demographic and agricultural characteristics of treatment households.

3. Study Population

For the purpose of this impact assessment, ETI-GB interventions were grouped into the following categories of project support:

- Irrigation and Land Development (ILD)
- Farm-to-Market Roads (FMR)
- Value Chain Development (VC)

Beneficiary households have received either a single type of support (e.g. ILD only, FMR only, VC only) or a combination of supports (e.g. ILD+FMR, ILD+VC, FMR+VC, or ILD+FMR+VC). As project monitoring systems recorded aggregated beneficiary numbers by component, but not consistently by overlapping intervention types, a consolidated beneficiary profiling dataset of households was used to estimate the distribution of beneficiaries across single and multiple intervention categories. The study population was therefore defined as beneficiary households listed in the official ETI-GB beneficiary records for ILD, FMR, and VC interventions with completed schemes and active production outcomes. In addition, households benefiting from multiple ETI-GB interventions were included in the sample in proportion to their estimated share in the overall beneficiary population, ensuring adequate representation of combined intervention effects.

4. Sampling Design and Sample Distribution

The impact assessment applies a probability-based, multi-stage sampling design to ensure that households within the defined study population have an equal chance of selection and that treatment–control comparisons are statistically and analytically robust. The sampling design is adapted to the structure and delivery modalities of ETI-GB interventions, which combine area-based infrastructure investments with household- and group-level support across multiple components.

At the first stage, primary sampling units are defined as completed or functionally active ETI-supported schemes and institutions. These include irrigation and land development schemes, farm-to-market road schemes and associated influence zones, and value chain institutions such as Village Agriculture Cooperatives. Primary sampling units are selected using probability proportional to size, based on official ETI-GB records of completed infrastructure schemes and supported value chain initiatives across districts. For value chain interventions, where multiple commodities are supported, disproportionate sampling of primary sampling units is applied to ensure adequate representation of major commodities.

At the second stage, households are selected as ultimate sampling units using a stratified approach. Households are first stratified by intervention exposure, including single-component support and combinations of irrigation and land development, farm-to-market roads, and value chain interventions. Within each intervention stratum, proportional allocation is applied at the district level to ensure geographic representation across Astore, Diamer, Ghanche, Ghizer, Hunza, Skardu, Shigar, Nagar, Gilgit, and Kharmang. Households are then randomly selected from official beneficiary lists within each district, thereby minimizing selection bias and ensuring that each eligible household has an equal probability of inclusion. The distribution of sampled treatment households across districts and intervention exposure categories is presented in Table 1.

District	Only ILD		Only FMR		Only VC		ILD+FMR		ILD+VC		FMR+VC		ILD+FMR+VC		Total	
	N	%	N	%	N	%	N	%	N	%	N	%	N	%	N	%
Astore	25	17.48	44	16.60	27	8.97	13	30.95	0	0.00	6	27.27	2	9.09	117	14.55
Diamer	58	40.56	63	23.77	35	11.63	4	9.52	4	44.44	1	4.55	9	40.91	174	21.64
Ghanche	25	17.48	58	21.89	31	10.30	17	40.48	0	0.00	7	31.82	2	9.09	140	17.41
Ghizer	35	24.48	96	36.23	30	9.97	7	16.67	0	0.00	1	4.55	0	0.00	168	20.90
Gilgit	0	0.00	0	0.00	18	5.98	0	0.00	2	22.22	0	0.00	4	18.18	24	2.99
Hunza	0	0.00	0	0.00	46	15.28	0	0.00	2	22.22	1	4.55	2	9.09	51	6.34
Kharmang	0	0.00	0	0.00	17	5.65	0	0.00	0	0.00	0	0.00	0	0.00	17	2.11
Nagar	0	0.00	1	0.38	32	10.63	0	0.00	0	0.00	0	0.00	0	0.00	33	4.10
Shigar	0	0.00	3	1.13	27	8.97	0	0.00	1	11.11	3	13.64	1	4.55	35	4.35
Skardu	0	0	0	0.00	38	12.62	1	2.38	0	0.00	3	13.64	2	9.09	45	5.60
Total	143	100	265	100	301	100	42	100	9	100	22	100	22	100	804	100

For the control group, households are selected from villages within the same districts where no ETI-GB interventions are implemented. Control villages are purposively identified to match treatment villages in terms of agro-ecological conditions, livelihood systems, and market access characteristics. Within selected control villages, households are randomly sampled using complete household listings. A consolidated overview of treatment and control samples by district is provided in Table 2. This summary demonstrates balance in geographic representation between the two groups and supports the validity of subsequent treatment–control comparisons across the outcome indicators analyzed in the assessment.

Table 2 Summary of Sampling Distribution				
District Name	Treatment Group		Control Group	
	N	(%)	N	(%)
Astore	117	14.55	107	13.61
Diamer	174	21.64	174	22.14
Ghanche	140	17.41	139	17.68
Shigar	35	4.35	33	4.20
Ghizer	168	20.90	164	20.87
Gilgit	24	2.99	22	2.80
Nagar	33	4.10	32	4.07
Hunza	51	6.34	48	6.11
Skardu	45	5.60	51	6.49
Kharmang	17	2.11	16	2.04
Total	804	100.00	786	100.00

5. Data Collection Methods

The impact assessment employed a mixed-methods data collection approach that integrates quantitative and qualitative tools to capture both measurable outcomes and contextual, behavioral, and institutional dimensions of ETI-GB interventions. Data collection was implemented across all sampled districts in accordance with the methodology approved in the inception report and is designed to support triangulation across evidence streams.

Quantitative data were collected through a structured household survey administered to treatment and control households. The survey captured information on household demographics, landholding and land use, access to irrigation, agricultural production and yields, livestock ownership, participation in value chain activities, enterprise engagement, market access, income sources, expenditure patterns, savings behavior, food security, health, and education outcomes. A separate women-focused questionnaire was administered within sampled households to capture gender-specific information, including dietary diversity, food consumption practices, access to health services, women's workload, and participation in economic activities. Additional modules on agriculture, livestock, and enterprise activities collected detailed information on inputs, practices, costs, revenues, and constraints to adoption. These data formed the primary basis for treatment–control comparisons and disaggregated analysis by intervention exposure.

Qualitative data was collected to complement and contextualize the household survey findings and to explain observed patterns and variations across districts and intervention types. Ten Focus group discussions were conducted in selected project locations representing different districts and intervention combinations. Focus Group Discussions were conducted to examine changes in agricultural production, income and livelihoods, market access and road usability, access to services, gender participation, collective action, institutional arrangements, and perceptions of sustainability. Semi-structured discussion guides were used to ensure consistency across sites while allowing participants to articulate their experiences in their own terms.

Eight KIIs were conducted with representatives of relevant government departments and implementing partners, including the Comms & Works Department, Planning and Development Department, Agriculture Extension, Irrigation and Water Management, Cooperative Department, Board of Revenue, and implementing organizations such as AKRSP and GBRSP. These interviews explored institutional roles, coordination mechanisms, implementation processes, policy alignment,

operational constraints, and prospects for sustainability and scale-up. KIIs provided insights into how ETI-GB interventions are embedded within provincial development systems and how institutional factors influence outcomes and long-term viability.

Field observations were undertaken in all visited locations to document the physical condition and functionality of irrigation schemes, command areas, farm-to-market roads, nurseries, orchards, enterprise units, and related infrastructure. Observations also covered water distribution practices, operation and maintenance arrangements, exposure to environmental hazards, accessibility to markets and services, and the activity level of community institutions. These observations were systematically recorded and used to verify and contextualize information obtained through surveys, FGDs, and KIIs. Together, the household survey, FGDs, KIIs, MSC stories, and field observations form a coherent body of evidence that enables triangulation across data sources. This integrated approach strengthens the robustness of findings, supports causal interpretation, and ensure that quantitative results are interpreted in light of local realities, institutional dynamics, and implementation conditions.

Chapter Three: Key Findings

1. Income & Changes in Income

This section examines the extent to which ETI-GB increased household income relative to non-beneficiaries, addressing the core impact question of whether project-supported households achieved measurably higher income outcomes than comparable households not receiving ETI-GB support. The analysis is explicitly aligned with the programme's log-framework, which targets that at least 50 percent of beneficiary households and value chain operators increase their agricultural income by a minimum of 25 percent. To answer this question, the section combines treatment-control comparisons with distributional analysis of income gains among beneficiaries, moving beyond descriptive reporting to assess both relative income performance and depth of income change. Evidence is synthesized across livelihood structure, incidence of income growth, magnitude and distribution of income gains, and the pathways through which income increases were achieved. Together, these analyses establish whether ETI-GB not only raised incomes among beneficiaries but did so to a greater extent than observed among non-beneficiary households operating under similar regional and economic conditions.

Structural Shifts in Livelihoods and Income Sources

An important early income-related effect of ETI-GB interventions is reflected in changes in how households earn their income, rather than only how much income they earn. Comparison of livelihood structures between treatment and control households reveals a clear shift in the composition of primary income sources among beneficiary households, indicating a transition toward more productive and locally anchored livelihood strategies

Table 3 Primary Sources of Households Income

Income Source	Categories	Treatment (%)	Control (%)
Agriculture & Livestock	Agricultural worker, Farming, Livestock breeding	30.60	16.40
Wage Employment	Civil servant, Private sector employee	38.70	46.70
Trade & Small Businesses	Trade/other small businesses, Handicraft	11.70	15.80
Construction & Manual Labor	Construction worker	10.70	8.10
Others	Pension/other benefits, Aid/assistance, & wood products, Other	8.30	13
Total		100	100

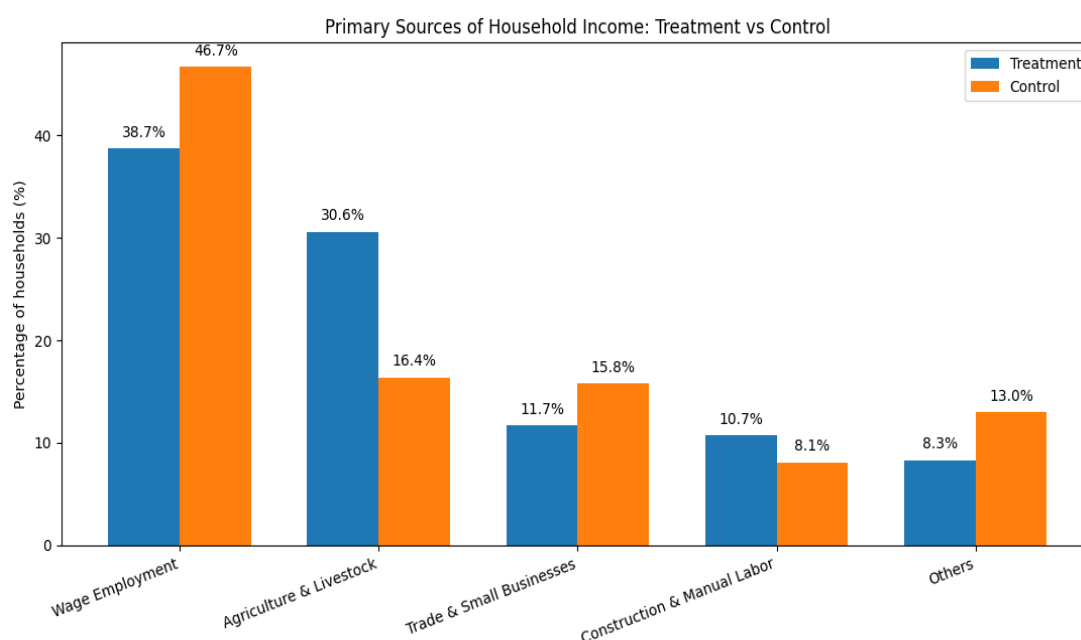


Figure 2 Primary Sources of Household Income

As shown in Table 3, treatment households are substantially more engaged in agriculture- and livestock-based livelihoods, with 30.6 percent reporting these activities as their primary income source, compared to 16.4 percent among control households. This nearly fourteen-percentage-point difference suggests that ETI-GB interventions have strengthened households' capacity and incentives to rely on land-based productive activities, consistent with the programme's causal pathway linking irrigation, land development, and agricultural commercialization to income generation. In contrast, reliance on wage employment remains higher among control households (46.7 percent) than treatment households (38.7 percent), indicating that in the absence of ETI support, households are more dependent on salaried or casual labor opportunities. Engagement in trade and small businesses is also more prevalent among control households (15.8 percent) than treatment households (11.7 percent), suggesting that enterprise activities are more commonly pursued as a primary livelihood where productivity-enhancing agricultural investments are not present.

Notably, treatment households show lower dependence on transfers and other non-productive income sources, including pensions, aid, and wood-based activities (8.3 percent, compared to 13.0 percent among control households). This reduction points to a shift away from dependency-oriented income sources toward more self-generated and sustainable livelihoods. At the same time, treatment households continue to diversify through construction and manual labor (10.7 percent), indicating that agricultural engagement under ETI-GB complements rather than fully replaces other income strategies.

These patterns indicate that ETI-GB interventions contributed to a structural reorientation of household livelihoods, favoring agriculture-based and productive income sources over wage dependence and transfers. This shift represents a foundational income pathway through which subsequent income growth, stability, and resilience outcomes observed in later sections were enabled.

Incidence of Income Change

This subsection assesses the incidence component of the logframe income indicator with a specific focus on agricultural income, examining whether a sufficient proportion of households experienced income increases consistently with the target of at least a 25 percent rise over the baseline annual agricultural income of PKR 45,000. ETI-GB interventions were designed to raise agriculture-based earnings through an integrated package of ILD, FMR, VC, and enterprise development. These interventions aimed to enhance productive capacity, reduce structural constraints in agricultural marketing, and strengthen households' ability to generate sustained income from farming activities. As such, observed income outcomes reflect the combined and reinforcing effects of these interventions rather than the influence of any single component.

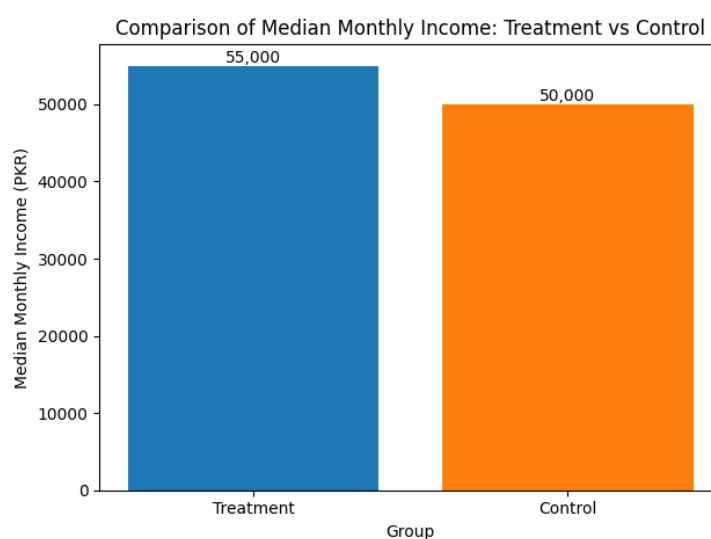


Figure 3 Median Monthly Income

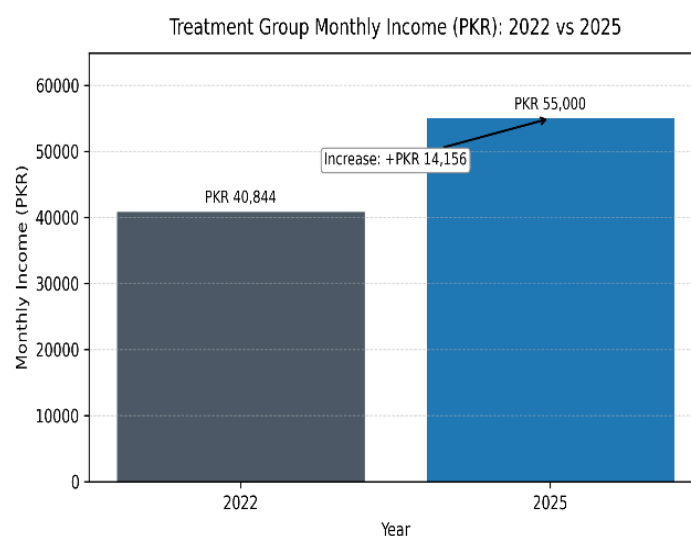


Figure 4 Monthly Income (PKR)

Figure 3 provides important contextual evidence by comparing mean monthly household income between treatment and control groups at the time of the survey. Treatment households reported a higher average (median) monthly income of PKR 55,000, compared to PKR 50,000 among control households, representing an income differential of PKR 5,000 (10 percent) in favor of beneficiaries.

While this overall income gap indicates a general welfare advantage associated with project participation, its relevance for the logframe is most clearly understood through the structure and composition of income—particularly the contribution of agriculture.

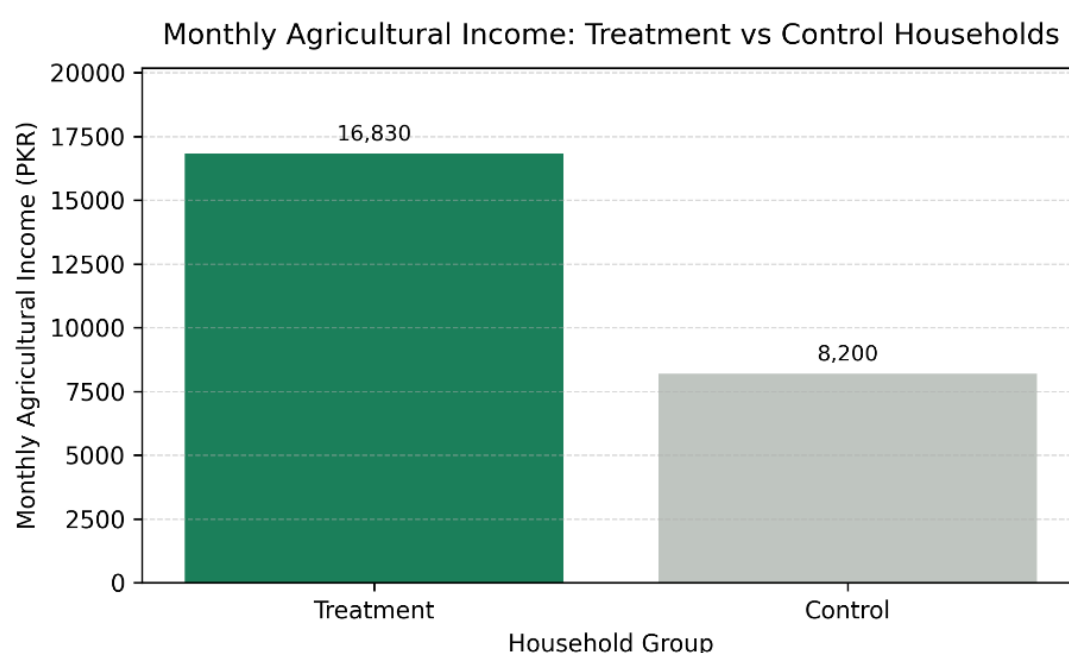


Figure 5 Monthly Agricultural Income of Treatment and Control Groups

Treatment households recorded substantially higher agricultural income levels than control households (Figure 5). Mean monthly agricultural income among treatment households was PKR 16,830, compared to PKR 8,200 among control households. On an annual basis, this corresponds to PKR 201,960 for treatment households, nearly double the PKR 98,400 recorded for control households.

Relative to the baseline annual agricultural income of PKR 45,000, agricultural earnings among treatment households increased by over 348 percent, far exceeding the log-frame target of a minimum 25 percent increase. While control households also experienced growth in agricultural income relative to the baseline, reflecting broader sectoral and economic trends, the additional income premium observed among treatment households—approximately a 10 percent higher annual agricultural income compared to control households—can be attributed to ETI-GB interventions.

This treatment–control differential provides strong evidence of attribution. Given that both groups operate under similar agro-ecological and market conditions, the markedly higher agricultural income among beneficiary households reflects the contribution of ETI-GB investments in improved irrigation, expanded productive land, enhanced market access, and value-chain integration. The results indicate not only higher absolute agricultural income levels but also a strengthened role of agriculture as a viable and remunerative livelihood strategy among treatment households.

These income gains are strongly validated by KIIs with Agriculture Extension, AKRSP, and cooperative representatives, who highlighted that ETI-GB interventions enabled a transition from subsistence farming to market-oriented production through expanded land access, improved post-harvest handling, and stronger market linkages.

Evidence from FGDs across ETI-GB intervention districts consistently confirms that agricultural income increases were substantial, sustained, and transformative rather than marginal. Participants repeatedly described ETI interventions—particularly irrigation and land development, improved market access, and value-chain support—as having shifted agriculture from subsistence activity to a primary source of cash income. Communities emphasized that income growth was sufficiently large to change household economic behavior, reduce reliance on wage labor, and enable regular engagement with markets. Participants reported that land previously considered barren or uneconomical was brought under cultivation, leading to sharp increases in production and sales. In Dahimal Ghizer, one participant stated,

“If I talk about myself personally, before this (ETI) I was buying PKR 100,000 worth of fodder annually and now I even sell fodder to others and hence my income increase significantly”.

Similar accounts from Astore and Diamer emphasized that agricultural income increased multiple times following irrigation development, with one participant in Fina Astore noting,

“due to ETI-interventions, my agricultural income has increased more than threefold, and overall living standards have improved”.

Value-chain support and collective marketing further amplified income gains, particularly in fruit- and vegetable-producing areas. FGDs in Hunza, Skardu, and Kharmang highlighted dramatic improvements in prices and earnings due to improved processing, grading, and collective sales through Village Agriculture Cooperatives. A cooperative member from Skardu explained:

“Earlier apricots were hardly sold at PKR 20–25 per kilo. Through the cooperative, we now sell at PKR 150–150 per kilo. This alone doubled our household income.”

Another participant in Passu Hunza added:

“.....through trainings provided by ETI we have been able to improve our crop production and it has helped in marketing and sales of produce in the market. These interventions have helped to increase the income of each beneficiary household and women have been enabled to earn own their own.

Distribution of Income Gains Among Beneficiaries

This subsection directly assesses performance against the logframe indicator requiring that at least 50 percent of target households increase their agricultural income by a minimum of 25 percent, using the distribution and magnitude of income gains among beneficiary households. Beyond whether household incomes increased, it is important to assess the magnitude and distribution of income gains among ETI-GB beneficiary households in order to understand the depth of the programme’s income effects. Table 7 and Figure 2 present the distribution of income changes across beneficiary households, distinguishing between modest, significant, and very large increases. This analysis provides insight into whether income improvements were marginal or transformative and helps clarify the extent to which ETI-GB interventions enabled households to move onto higher income growth trajectories.

Intervention Category	Income Increased (Yes %)	Modest Increase (20–49%)	Significant Increase (50–99%)	Very Large Increase (≥100%)	Very Small Increase (<20%)
HH	86.61	30.35	48.70	10.84	10.12

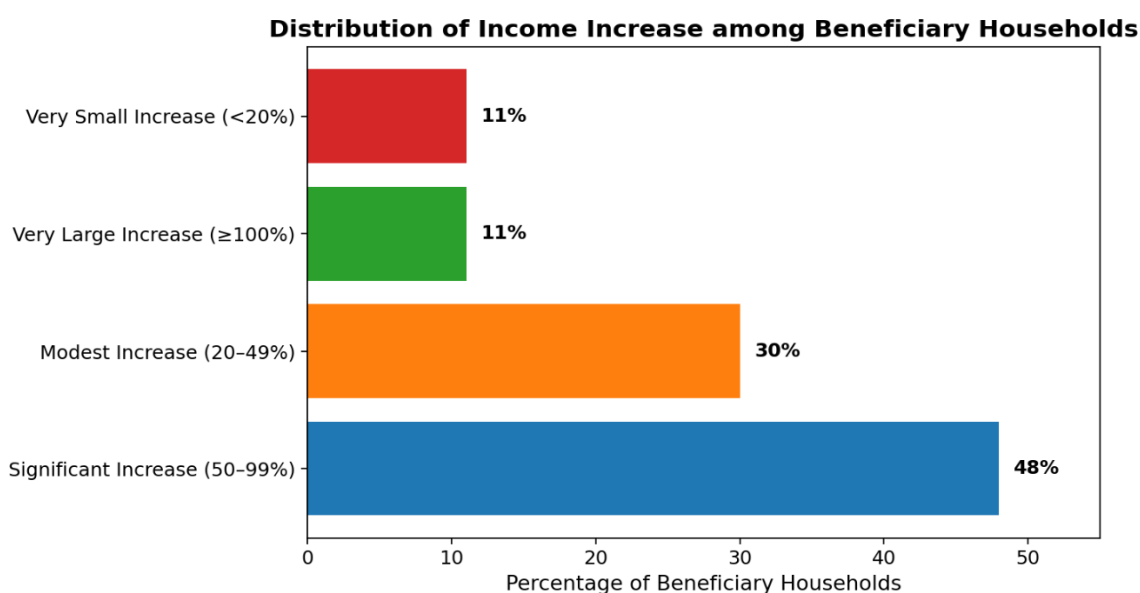


Figure 6 Distribution of income among beneficiary households

Table 4 and Figure 6 illustrate that a large majority of beneficiary households (86.61 percent) reported an increase in income following the intervention, confirming that income gains were widespread rather than confined to a small subset of participants. More importantly, the distribution of income gains indicates that improvements were substantial rather than incremental. Nearly half of beneficiary households (48.70 percent) experienced a significant income increase of 50–99 percent, while an additional 10.84 percent reported very large income gains of 100 percent or more, effectively doubling their income. Together, these results indicate that close to six in ten households achieved income growth exceeding 50 percent, reflecting transformative income effects for a sizeable share of beneficiaries. At the same time, 30.35 percent of households reported modest income increases (20–49 percent), suggesting that the intervention also generated positive but more moderate gains across a broad segment of participants. Only a relatively small proportion of households (10.12 percent) experienced very small income increases below 20 percent, indicating limited income response among a minority of beneficiaries.

These findings demonstrate that ETI-GB interventions did not merely raise incomes at the margin but enabled a significant share of households to achieve large and sustained income gains. With more than half of beneficiary households experiencing income increases well above the 25 percent threshold, the evidence indicates that ETI-GB meets and exceeds the income-related log-frame target, particularly through significant and very large income gains among beneficiaries

Pathways Through Which Income Gains Were Achieved

Understanding how income gains were realized is essential for interpreting the nature and sustainability of ETI-GB's income effects. This subsection examines the key pathways through which beneficiary households experienced income increases, drawing on household-reported factors and comparing responses between treatment and control groups. The analysis helps distinguish between production-driven, market-driven, and other mechanisms, thereby clarifying how ETI-GB interventions translated into higher household incomes.

Table 5 Key Factors Contributing to Increase in income		
Contributing Factor	Treatment (%)	Control (%)
Increase in irrigated area	76.80	63.95
Increase in volumes produced	52.00	35.31
Better sale prices due to quality	30.40	23.81
Increase in volumes sold	16.00	14.97
Better sale prices due to bargaining / market access	16.80	12.93
Able to sell when prices are high	7.20	6.12
Other factors	8.00	8.16

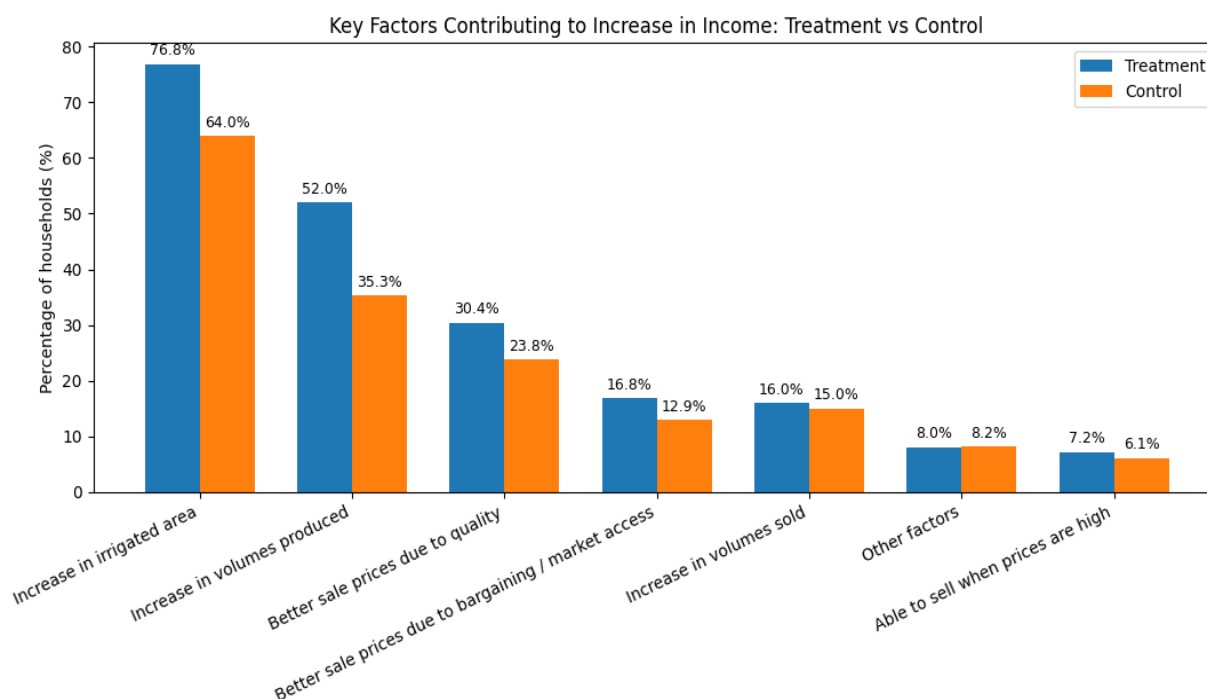


Figure 7 Key factors contributing to income increased.

Table 5 and figure 7 highlight clear differences between treatment and control households in the factors reported as contributing to income increases, providing insight into the mechanisms through which ETI-GB interventions influenced household earnings. The most prominent pathway is the increase in irrigated area, reported by 76.80 percent of treatment households compared to 63.95 percent of control households. This difference underscores the central role of irrigation and land development investments in expanding productive capacity, stabilizing agricultural production, and enabling surplus generation among beneficiary households. Increases in volumes produced represent the second most frequently cited factor, reported by 52 percent of treatment households versus 35.31 percent of control households. This substantial gap suggests that ETI-GB interventions enabled households to move beyond subsistence-level production toward higher output levels, reinforcing income gains through productivity improvements rather than reliance on price effects alone. Market-related pathways also contributed to income growth, though to a lesser extent than production-side factors. A higher proportion of treatment households reported better sale prices due to improved quality (30.4 percent) compared to control households (23.81 percent), indicating that project-supported improvements in production practices and post-harvest handling enhanced market returns. Similarly, treatment households were more likely to cite better sale prices due to bargaining power and market access (16.8 percent vs. 12.93 percent) and increased volumes sold (16 percent vs. 14.97

percent), reflecting the role of improved connectivity and commercialization in translating production gains into cash income. The findings demonstrate that ETI-GB's income effects were achieved predominantly through expansion of productive capacity and increased output, complemented by improvements in product quality and market engagement. This pattern is consistent with the project's theory of change, which emphasizes irrigation-led productivity growth and gradual market integration as the foundation for sustainable household income improvements.

FGD findings strongly corroborate the factors reported in Table 5 as key contributors to household income increases among ETI-GB beneficiaries. The most dominant driver—expansion of irrigated area—was consistently emphasized in FGDs particularly in Diamer and Ghizer. Participants reported that irrigation development enabled cultivation of previously unused land, increased cropping intensity, and reduced production risk, directly translating into higher income. As one participant noted in Diamer: “... when water reached our land, agriculture became a source of cash income, not just food.”

This qualitative evidence supports the high proportion of treatment households attributing income growth to expanded irrigation. The quality data also provide strong support for increased production volumes as a major income pathway. Participants across Nagar, Astore, and Ghanche described substantial increases in output of vegetables, cereals, and fruits, allowing households to sell surplus regularly. A farmer from Nagar explained: “... earlier we produced only for home use. Now production has increased so much that selling has become normal.”

Evidence for better sale prices due to improved quality is particularly strong in areas with value-chain support. Participants from Hunza, Skardu, and Kharmang described improved grading, drying, and packaging—especially for apricots—which enabled access to higher-value markets. One cooperative member stated: “... *before, apricots sold very cheaply. After training, we sell at much higher prices, and income has increased a lot.*”

This explains why quality-related price improvements were more frequently reported by treatment households. Furthermore, FGDs also substantiate the role of increased volumes sold and improved bargaining through better market access, mainly as secondary effects following higher production. Participants highlighted reduced transport costs, traders reaching villages, and stronger bargaining power through Village Agriculture Cooperatives.

The evidences from both quantitative and qualitative data demonstrate that ETI-GB increased household income to a measurable and meaningful extent compared to non-beneficiaries, while also enabling a large share of beneficiary households to achieve income gains well above the programme's minimum threshold. Treatment households report a higher average monthly income (PKR 55,000) than control households (PKR 50,000), resulting in an income differential of approximately PKR 5000, or 10 percent, in favor of ETI-GB beneficiaries. This treatment–control gap indicates that income gains among beneficiaries cannot be attributed solely to broader economic or contextual trends affecting both groups, but reflect the additional contribution of ETI-GB interventions. Beyond differences in average income levels, the distribution of income gains among beneficiaries provides stronger evidence of impact. A large majority of beneficiary households (86.6 percent) reported an increase in income, and—critically—nearly 60 percent experienced income gains exceeding 50 percent, with 10.8 percent doubling their income or more. These results indicate that ETI-GB did not merely raise incomes marginally relative to non-beneficiaries, but enabled a substantial share of households to achieve transformative income growth, far exceeding the 25 percent benchmark set in the logframe.

The pathways analysis further clarifies why beneficiary households outperformed non-beneficiaries. Treatment households were significantly more likely than control households to report income gains driven by expanded irrigated area (76.8% vs. 64.0%), increased production volumes (52.0% vs. 35.3%), and improved sale prices linked to quality and market access. These differences confirm that ETI-GB's integrated investments in irrigation, land development, and market connectivity strengthened productive capacity and commercialization outcomes in ways not observed to the same extent among non-beneficiaries.

Taken together, the findings show that ETI-GB increased household income both in relative terms (higher average incomes than non-beneficiaries) and in absolute terms (large and widespread income gains among beneficiaries). The programme therefore met and exceeded its income objective, with impacts driven by structurally grounded mechanisms—expanded production, higher productivity, and improved market engagement—rather than short-term or isolated effects. These income gains provide a robust foundation for the subsequent poverty reduction, food security, and resilience outcomes examined in later sections of the report.

2. Poverty Analysis

Poverty reduction is one of the key goals of ETI Interventions. This section assesses ETI's contribution to poverty reduction explicitly against the project's goal-level log-frame indicator on reduction of poor households, measured through the requirement of a 15 percent reduction in the number of poor households compared to the 2014 baseline. Using the nationally standardized Benazir Income Support Programme (BISP) Poverty Scorecard methodology, the analysis examines changes in poverty incidence among beneficiary households over time and relative to non-beneficiaries. Findings are synthesized across intervention types to assess whether observed welfare improvements meet the poverty reduction threshold specified in the results framework and to understand how different intervention combinations contributed to these outcomes. A poverty scorecard analysis was conducted (Table 6) for both the treatment and control households using the same methodology applied under Pakistan's Benazir Income Support Programme (BISP), ensuring national comparability and methodological rigor. The results were then benchmarked against the baseline poverty assessment carried out by ETI in 2022, allowing for a meaningful comparison of welfare changes over time.

The results (Table 6) of the poverty scorecard analysis reveal clear differences in household welfare between the Treatment and Control groups, as well as across the different categories of project interventions. At the overall level, the Treatment group demonstrates a slightly better welfare position compared to the Control group. A higher proportion of households in the Treatment group fall within the non-poor category (47.0% compared to 45.8% in the Control group), and fewer Treatment households fall into the chronically poor or extremely poor categories. Although the difference between the two groups is modest, it nonetheless suggests that project participation is associated with marginal improvements in household well-being. These improvements may be influenced by variations in the intensity and type of interventions received, which become more visible once the results are disaggregated by intervention type. The poverty headcount index (HCI) was calculated using the Poverty Score Card (PSC), classifying households in the transitory poor, chronically poor, and extremely poor categories as poor. Using this approach, the headcount index for the treatment group is 3.61 percent, which is substantially lower than the baseline poverty level of 34.60 percent and also lower than the control group headcount of 5.60 percent. This marked reduction relative to the baseline indicates a significant decline in poverty incidence among ETI beneficiary households over time, while the lower poverty level compared to the control group highlights the positive impact of

the intervention, suggesting that project-supported households were more effective in avoiding poverty or moving out of poverty than non-beneficiary households.

Table 6 Poverty Score Card				
Poverty Category	Treatment		Control	
	N	%	N	%
Non-poor	378	47.01	360	45.80
Transitory non-poor	332	41.29	324	41.22
Transitory vulnerable	65	8.08	58	7.38
Transitory poor	16	1.99	35	4.45
Chronically poor	11	1.37	7	0.89
Extremely poor	2	0.25	2	0.25
Headcount Index (%)	3.61		5.60	
Headcount index (Baseline-2014) (%)	34.60			

Table 7 Comparative Summary: Poverty Status Across Groups			
Poverty Category	Treatment Group (%)	Control Group (%)	2022 ETI-GB Beneficiaries (%)
Non-poor	47.01	45.80	8.00
Transitory non-poor	41.29	41.22	41.00
Transitory vulnerable	8.08	7.38	34.00
Transitory poor	1.99	4.45	9.00
Chronically poor	1.37	0.89	6.00
Extremely poor	0.25	0.25	1.00
Total Poor in %	3.61	5.60	16.00

As shown in Table 7, The comparison clearly demonstrates the poverty-reducing effect of ETI interventions. The treatment group exhibits a substantially lower poverty headcount index (3.61%) compared to the control group (5.60%), indicating that beneficiary households were less likely to fall into transitory, chronic, or extreme poverty. When compared with the 2022 BISP PSC results for ETI-GB beneficiaries, where 16 percent of households remain poor, the treatment group shows a much stronger poverty outcome, suggesting that intensive, asset-based interventions such as ILD generate deeper poverty reduction effects than broader beneficiary coverage alone. The control group's higher share of transitory poor households (4.45%), relative to the treatment group (1.99%), further highlights the intervention's role in preventing downward mobility. Overall, this triangulated comparison indicates that ETI support—particularly ILD-focused interventions—has been effective not only in reducing poverty incidence but also in shifting households into more secure, non-poor and transitory non-poor categories.

Poverty score across interventions

A more detailed examination across the various intervention groups highlights substantial gains for households that received targeted support (Table 8). The strongest improvements are observed among the VC and integrated intervention groups. For instance, the FMR-VC group shows the highest proportion of non-poor households at 63.6%, followed by the ILD FMR group (61.9%) and the Only VC beneficiaries (57.7%). These groups also exhibit negligible levels of transitory poor, chronic poor, or extreme poor households. This pattern indicates that interventions that combine infrastructure (such as farm-to-market roads) with livelihood development or value chain strengthening tend to generate

more robust and sustained poverty reduction outcomes. Such integrated approaches appear to improve household income opportunities, access to markets, and economic resilience, thereby lifting a substantial share of households out of poverty. Moderate improvements are also visible in other intervention groups, such as ILD VC, where more than half of the households qualify as non-poor and none fall into the poor categories. In contrast, the FMR-only group shows relatively weaker performance, with only 34.5% of households classified as non-poor. This suggests that road access alone may not sufficiently enhance household welfare unless accompanied by livelihood-enhancing components. The ILD-FMR-VC group presents a unique case with higher levels of vulnerability and a small but notable presence of chronically poor households. This may reflect underlying structural disadvantages, more challenging socioeconomic conditions, or the fact that households in this group may require a longer period to fully realize the benefits of multiple interventions.

The distribution of poverty across the intervention groups underscores several important insights. First, integrated interventions, particularly those combining infrastructure with value chain support demonstrate the strongest contribution to poverty reduction. Second, vulnerability persists among households with weaker access to diversified livelihood opportunities, such as those dependent on standalone interventions. Third, deep poverty remains relatively low across most groups, indicating that project activities have helped reduce chronic deprivation. Nonetheless, targeted follow-up may be required for certain segments, such as households in the ILD FMR VC group, where vulnerability and chronic poverty remain comparatively higher.

Table 8 Poverty Score Card across Interventions

Group	Non-poor (%)	Transitory non-poor (%)	Transitory vulnerable (%)	Transitory poor (%)	Chronically poor (%)	Extremely poor (%)
FMR	34.47	50.76	8.71	3.79	1.89	0.38
Only VC	57.70	34.10	6.89	0.98	0.33	0.00
FMR VC	63.64	31.82	4.55	0.00	0.00	0.00
ILD FMR VC	47.37	26.32	15.79	5.26	5.26	0.00
ILD FMR	61.90	30.95	7.14	0.00	0.00	0.00
ILD VC	55.56	44.44	0.00	0.00	0.00	0.00
ILD Only	40.79	45.39	9.21	1.32	2.63	0.66

Poverty scorecard results between 2022 and 2025

The comparison of poverty scorecard results between 2022 and 2025 across intervention groups (Table 9) reveals substantial and broad-based improvements in household welfare. All intervention categories show upward mobility, but the magnitude of change varies significantly depending on the type and intensity of support received. Across nearly all groups, households have moved decisively from poverty and vulnerability into the non-poor and transitory non-poor categories, indicating strong positive impacts attributed to project interventions. Households receiving Integrated Livelihood Development (ILD) only demonstrate noticeable improvements, with non-poor households rising from just 6% in 2022 to nearly 41% in 2025. Vulnerable and poor households sharply declined, reflecting the effectiveness of livelihood diversification measures. Farm-to-Market Road (FMR) beneficiaries show even stronger gains, with non-poor households increasing from 5% to over 34%, alongside significant reductions in transitory and chronic poverty levels. This indicates that improved market connectivity has contributed to enhanced income stability and welfare. The Value Chain (VC) intervention continues to be one of the most impactful components of the project. VC-only

households show a major transition: non-poor households almost doubled from 30% to 57.7%, while vulnerable and poor households dropped to negligible levels. This highlights how linking farmers with markets, improving product quality, and strengthening supply chains translates directly into measurable welfare gains. Even more striking improvements emerge among households benefiting from combined interventions. The ILD + FMR group shows one of the most dramatic shifts in welfare: non-poor households surged from 7% to 61.9%, while all categories of poverty (transitory, chronic, and extreme) effectively disappeared by 2025. This clearly demonstrates the synergy effect of combining improved infrastructure access with livelihood support. Similarly, households in the ILD + VC group experienced exceptional progress, moving from 0% non-poor in 2022 to 55.6% in 2025, while all poverty categories were eliminated. This suggests that combining income-generating interventions with institutional or livelihood strengthening produces a highly effective and sustainable welfare outcome.

Even for the group receiving the most complex combination of interventions (ILD + FMR + VC), welfare improvements are substantial. The non-poor category rose from 2% to 47.4%, and the share of chronically and extremely poor households dropped significantly. Although this group still displays slightly higher levels of vulnerability than other intervention bundles, the overall direction of change remains strongly positive. This may reflect that households in this category started from a more disadvantaged baseline and may require longer-term support to fully capitalize on the interventions.

Intervention Group	Year	Non-poor (%)	Transitory non-poor (%)	Transitory vulnerable (%)	Transitory poor (%)	Chronically poor (%)	Extremely poor (%)
ILD Only	2022	6.00	45.00	36.00	10.00	3.00	0.00
	2025	40.79	45.39	9.21	1.32	2.63	0.66
FMR	2022	5.00	44.00	28.00	13.00	9.00	1.00
	2025	34.47	50.76	8.71	3.79	1.89	0.38
VC Only	2022	30.00	44.00	19.00	6.00	1.00	0.00
	2025	57.70	34.10	6.89	0.98	0.33	0.00
ILD + FMR	2022	7.00	37.00	41.00	5.00	9.00	1.00
	2025	61.90	30.95	7.14	0.00	0.00	0.00
ILD + VC	2022	0.00	37.00	43.00	12.00	6.00	1.00
	2025	55.56	44.44	0.00	0.00	0.00	0.00
ILD + FMR + VC	2022	2.00	35.00	41.00	9.00	10.00	4.00
	2025	47.37	26.32	15.79	5.26	5.26	0.00

Qualitative evidence from the FGDs indicates that the income gains generated through ETI-GB interventions are translating into meaningful poverty-reduction effects, primarily by improving households' ability to meet basic needs, reducing economic vulnerability, and decreasing reliance on negative coping strategies. Participants across districts consistently described changes in livelihoods and living conditions that reflect improved economic security and reduced poverty-related stress. Households reported that increased agricultural income enabled them to meet a larger share of their food requirements from own production while generating surplus for sale. This shift reduced dependence on market purchases for staple foods and lowered exposure to price fluctuations, contributing to more stable consumption patterns. Such improvements in food self-sufficiency and cash availability are widely associated with reduced poverty and improved household welfare. FGDs further suggest a decline in reliance on casual wage labor and seasonal migration as primary livelihood

strategies. Participants described greater capacity to sustain livelihoods locally through farming and related activities, supported by improved irrigation, production, and market access. This transition away from precarious and low-paid labor reflects improved livelihood viability and reduced vulnerability—key dimensions of poverty reduction. Participants also reported increased ability to allocate income toward essential welfare needs, including education, health care, and housing improvements. These changes indicate that income gains were not limited to immediate consumption, but extended to investments that strengthen household well-being and long-term resilience. Such patterns are consistent with pathways through which income growth contributes to sustained poverty reduction.

Women’s participation in income-generating activities further reinforced these dynamics. FGDs indicate that income earned by women through processing, collective marketing, and small-scale agricultural activities contributed directly to household welfare and basic needs, enhancing the inclusiveness and stability of income gains.

The quantitative and qualitative evidences presented in this section demonstrates that ETI-GB has met and substantially exceeded the poverty reduction target specified in the project log-frame. Poverty scorecard analysis shows that the proportion of poor households among beneficiaries declined from 34.6 percent at baseline to 3.61 percent in 2025, representing a reduction of more than 30 percentage points, well above the 15 percent reduction threshold defined under the PDR indicator. Comparisons with control households further indicate that ETI-GB beneficiaries were less likely to experience transitory, chronic, or extreme poverty, strengthening confidence that observed poverty reduction is attributable, at least in part, to project interventions. Disaggregated results across intervention bundles highlight those integrated approaches—particularly those combining irrigation and land development, market connectivity, and value-chain support—generated the strongest and most sustained poverty reduction outcomes. Taken together, these findings confirm that ETI-GB achieved its intended poverty reduction objective at goal level, contributing meaningfully to improved welfare conditions in rural Gilgit-Baltistan.

3. Health and Food Security

This section assesses ETI-GB’s contribution to improved household health and food security against the nutrition-related output indicator CI 1.1.8 (households provided with targeted support to improve their nutrition) and its associated welfare outcomes. The analysis examines whether project-supported livelihood and nutrition-sensitive interventions translated into improved access to medical care, reduced food insecurity, and better child health and nutrition outcomes among beneficiary households. Findings are interpreted using treatment–control comparisons and beneficiary perceptions to assess progress toward improved household well-being in line with IFAD core outcome expectations.

Table 10 and the accompanying figures show that treatment households have better access to medical care than control households. A higher share of treatment households (64.8%) report being fully able to seek medical attention when needed, compared to 55.1% among control households. In contrast, control households more often seek care only when illness is severe, indicating delayed or constrained access. Financial barriers to healthcare are also lower among treatment households, with fewer reporting difficulty affording medicines or being unable to access care at all. Overall, the table and figure together indicate that project participation is associated with improved healthcare access and reduced affordability constraints, consistent with IFAD core outcomes on household resilience and well-being.

Table 10 Access to Medical Attention		
Access Category	Treatment (%)	Control (%)
Fully able to seek medical attention	64.80	55.10
Able only in serious illness	32.30	37.30
Access but affordability constrained	1.90	5.10
Not able to seek medical attention	0.70	2.50
Not relevant	0.20	—
Total	100	100

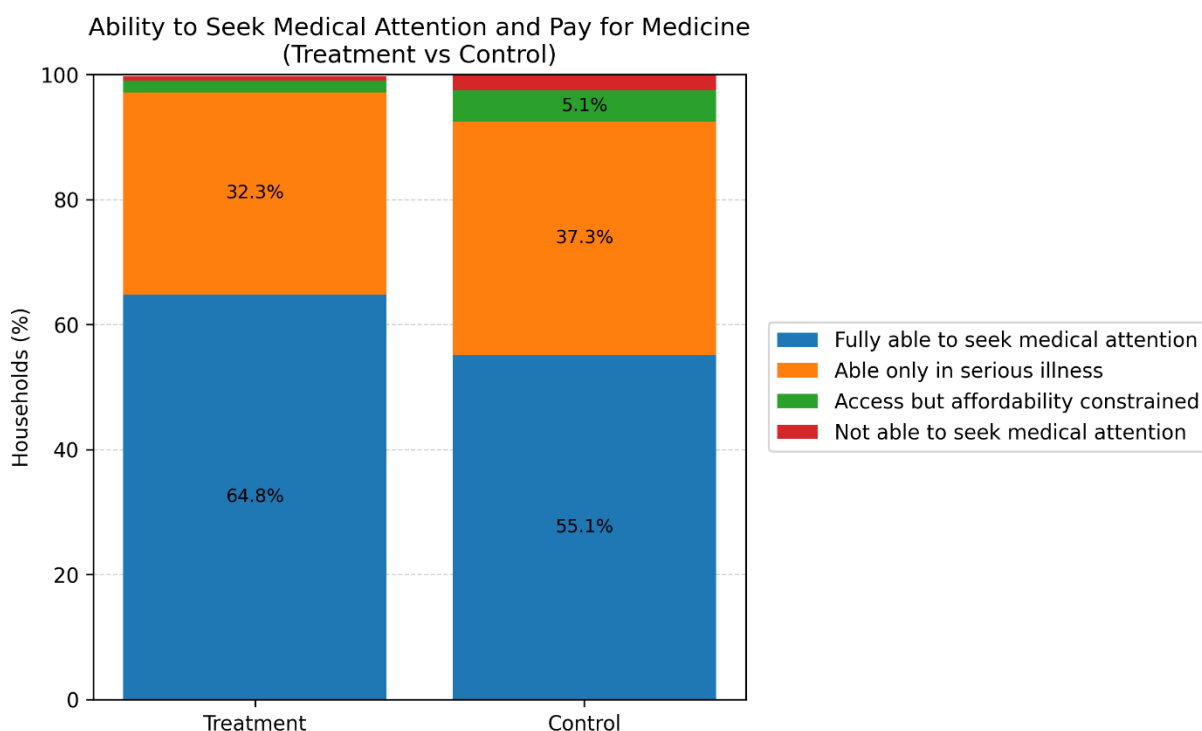


Figure 8 Ability of Households to Seek Medical Attention and Pay for Medicine

Figure 9 compares district-wise food insecurity between control and treatment households, excluding districts where no food insecurity was reported. Across all food-insecure districts (Gilgit, Ghanche, Diamer, Ghizer, and Nagar), the treatment group consistently reports lower levels of food insecurity than the control group. The largest reductions are observed in Ghanche and Diamer, followed by Ghizer and Nagar, indicating a positive association between project interventions and improved household food security outcomes.

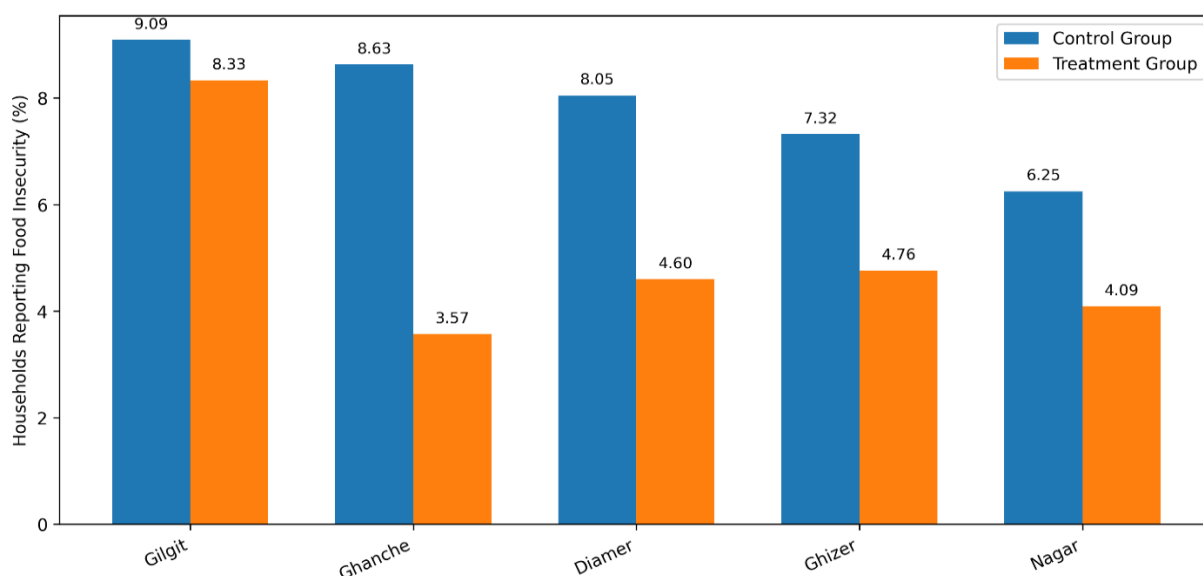


Figure 9 Household reported food insecurity across treatment and control groups

Responses indicate that the project is associated with notable improvements in child health and nutrition across intervention types (Table 11 and Figure 10). Overall, two-thirds of households (66.9%) report some improvement, while 14.7% report major improvement, and fewer than one-fifth observe no change. The strongest outcomes are observed among households receiving combined interventions, particularly ILD+FMR+VC, where nearly 47% report major improvements. In contrast, single-intervention households mainly report incremental (“some”) improvements, suggesting that integrated, multi-component interventions are more effective in generating substantial gains in child health and nutrition.

Table 11 Perceived Change in Child Health and Nutrition Since Project Start				
Intervention Type	Major Improvement (%)	Some Improvement (%)	No Change (%)	Total (%)
Percentage Of HH	14.68	66.87	18.45	100

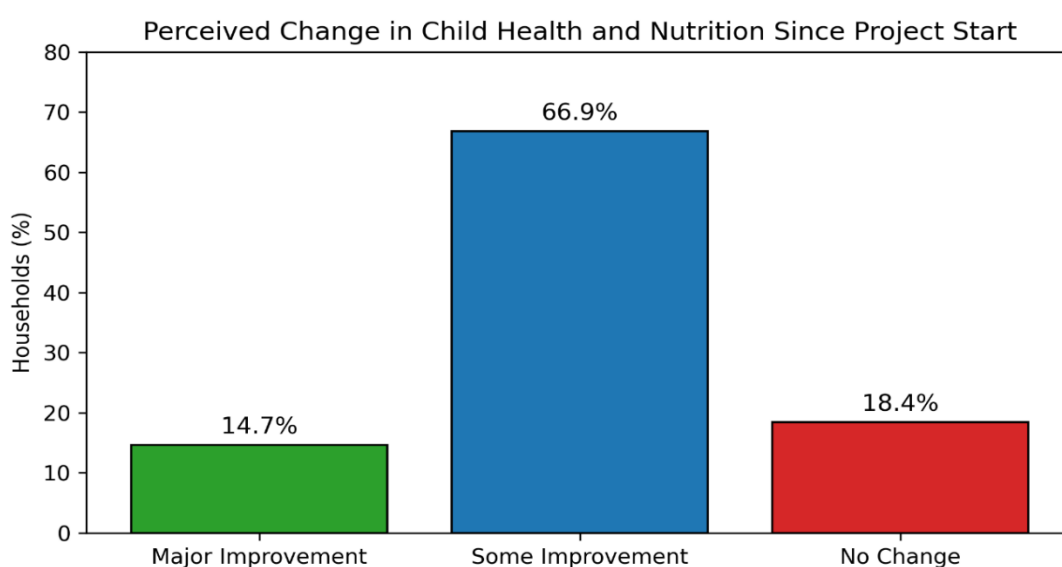


Figure 10 perceived change in child health and nutrition

Table 11 and the accompanying figure 6 present beneficiary households' perceptions of changes in child health and nutrition since the start of the project. The results indicate predominantly positive outcomes, with a large majority of households reporting improvements over the project period. Overall, 81.6 percent of households perceive an improvement in child health and nutrition, comprising 66.9 percent reporting some improvement and 14.7 percent reporting major improvement. This suggests that while improvements have been gradual for most households, a meaningful proportion have experienced substantial gains. These perceptions are consistent with the project's emphasis on improving food security, nutrition awareness, and household incomes, which together contribute to better dietary intake and child well-being. Only 18.4 percent of households report no change in child health and nutrition, indicating that stagnation is limited to a minority of beneficiaries. Overall, the findings provide strong supporting evidence that project-supported interventions are positively influencing child health and nutrition outcomes, as perceived by beneficiary households. These results align with Core Outcome Indicators related to nutrition and household well-being and reinforce the conclusion that integrated livelihood, income, and nutrition-related activities contribute to improved child nutrition outcomes over time.

KIIs with AKRSP and P&DD confirm that women's participation in greenhouse farming, value-chain activities, and nutrition awareness initiatives contributed to these outcomes, while also emphasizing that cultural norms in districts such as Diامر continue to limit women's direct engagement, underscoring the need for context-specific approaches

FGD participants across intervention areas reported noticeable improvements in the quality and diversity of household diets, which they associated with increased local food production and better market access. In Surmo village of District Ghanche, one participant highlighted a clear shift away from traditional, monotonous diets toward more diversified meals:

"The quality of our food has improved. Earlier, people mostly relied on namkeen chai and local roti, but now households are growing different vegetables and preparing a variety of foods. This has brought a visible change in our diet." (FGD participant, Surmo, Ghanche)

Improved physical connectivity was also identified as an important enabler of food diversity and nutritional intake. Participants explained that better access roads facilitated the availability of fresh vegetables and food items, benefiting both producers and consumers. A participant from Fina village in Astore noted:

"With improved access roads, fresh vegetables and food items reach markets and households more easily. As a result, food diversity and nutritional intake have improved for both men and women." (FGD participant, Fina, Astore)

At the same time, FGDs revealed growing awareness of nutrition and health, alongside recognition of emerging challenges. In Passu (Hunza), a participant emphasized increased understanding of the relationship between diet and health, while also expressing concern over rising consumption of processed foods, particularly among children:

"There is growing awareness about nutrition and how food affects our health and diseases. However, the consumption of junk food has also increased, especially among children. I think we need more awareness so people rely more on healthy, locally produced food." (FGD participant, Passu, Hunza)

Composite Dietary Diversity Index (DDS)

The Women's Dietary Diversity Index (MDD-W) was constructed following the FAO-recommended approach using ten standard food groups, including grains and tubers, pulses, nuts and seeds, milk and milk products, meat/fish/poultry, eggs, dark green leafy vegetables, vitamin A-rich fruits and vegetables, other vegetables, and other fruits. For each food group, women who reported consumption were assigned a value of 1, while non-consumption was assigned 0. The index therefore ranges from 0 to 10, with higher scores indicating greater dietary diversity and improved nutritional adequacy. The composite index presented in the figure reflects the average number of food groups consumed by women in treatment and control households, providing a summary measure of dietary quality and diversity.

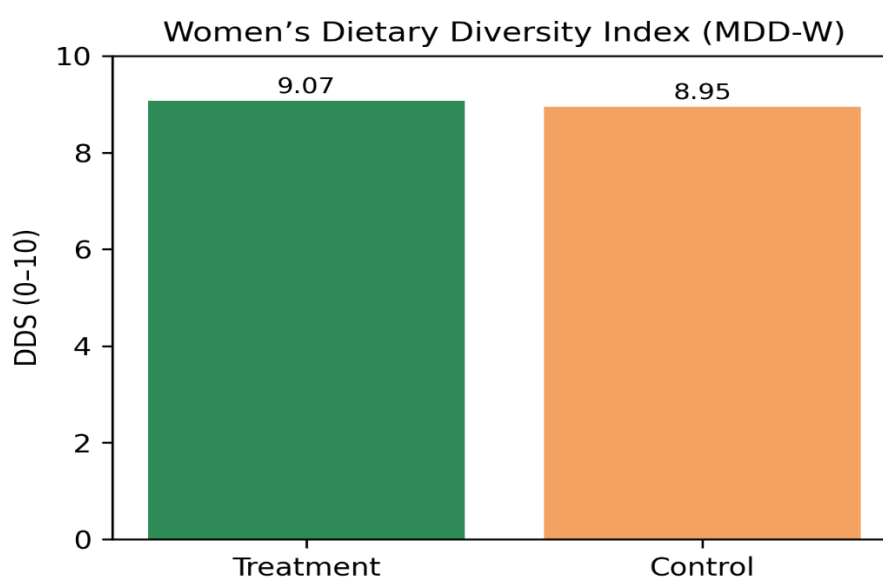


Figure 11 Composite Dietary Diversity Index

The results indicate (Figure 11) a higher average DDS among treatment households (9.07) compared to control households (8.95), suggesting that ETI beneficiaries have more diverse diets overall. This difference is driven mainly by higher frequent consumption among treatment households of animal-source foods (meat/fish, eggs, milk) and pulses, alongside consistently high consumption of vegetables and fruits in both groups. Overall, the DDS comparison provides supportive evidence that ETI participation is associated with improved dietary diversity, consistent with the project's nutrition-sensitive pathway linking livelihood gains and improved food access to better household diets.

The evidence indicates that ETI-GB has achieved meaningful progress toward the nutrition-related output indicator CI 1.1.8 (households provided with targeted support to improve their nutrition) through integrated livelihood and nutrition-sensitive interventions. Treatment households demonstrate stronger outcomes across multiple dimensions of household well-being, including healthcare access, food security, dietary diversity, and child nutrition. Access to healthcare is notably better among beneficiary households, with 64.8 percent reporting being fully able to seek medical attention when needed, compared to 55.1 percent among control households, alongside fewer reports of affordability and access constraints. Food security outcomes similarly favor treatment households across all food-insecure districts, indicating improved and more stable access to food associated with project participation. Improvements in child health and nutrition are widely reported,

with 81.6 percent of beneficiary households perceiving positive change since the start of the project, including 14.7 percent reporting major improvement. Stronger outcomes among households receiving integrated intervention packages (ILD+FMR+VC) highlight the effectiveness of combined livelihood and market-access support in generating nutrition gains. Dietary diversity outcomes further reinforce these findings, with treatment households exhibiting a higher average Composite Dietary Diversity Score (3.96) than control households (3.51), driven by more frequent consumption of nutrient-dense foods. Qualitative evidence complements these results by illustrating improved food quality, increased consumption of vegetables and diverse foods, and growing awareness of nutrition and health practices at the household level. Together, the findings demonstrate that CI 1.1.8 was achieved through a nutrition-sensitive pathway linking increased agricultural production, higher household income, improved market access, and enhanced food choices, resulting in measurable improvements in household health and food security in rural Gilgit-Baltistan.

4. Knowledge Attitude and Practices (KAP)

This section evaluates ETI-GB's impact against the outcome-level nutrition and behavior indicators CI 1.2.8 (women reporting Minimum Dietary Diversity – MDD-W) and CI 1.2.9 (households with improved Knowledge, Attitudes and Practices – KAP). The analysis focuses on whether targeted nutrition support and awareness activities contributed to improved dietary diversity among women, enhanced micronutrient knowledge, and stronger food hygiene behaviors among beneficiary households. Findings are based on composite indices and treatment–control comparisons to assess measurable progress against these outcome indicators.

Intake of micronutrients

The results indicate (Table 12) generally high adoption of basic nutrition practices, particularly salt use, across both treatment and control groups. Almost all households reported using salt in the preparation of their main meal, with a slightly higher prevalence in the treatment group (96.9%) compared to the control group (94.3%). Use of iodized salt is also widespread in both groups (around 69%), suggesting good market availability; however, a substantial proportion of respondents—over one quarter in both groups—were unable to identify the type of salt used, pointing to persistent gaps in nutrition knowledge.

Table 12 Comparison of Micronutrient-Related Practices and Knowledge

Indicator	Treatment Group (%)	Control Group (%)	Difference (T – C) (% pts)
Used salt in last cooked meal (Yes)	96.89	94.27	2.62
Use of iodized salt	69.03	68.83	0.20
Do not know type of salt used	28.11	27.26	0.85
Heard about iron deficiency (anemia)	37.31	34.99	2.32
Correctly identified lack of iron as cause of anemia	77.00	66.18	10.82
Believe vitamin A deficiency is serious	34.33	29.01	5.32
Do not know seriousness of vitamin A deficiency	51.62	56.87	–5.25

Awareness of iron deficiency (anemia) remains moderate overall but is consistently higher among treatment households. Importantly, the treatment group demonstrates a stronger understanding of the causes of anemia: 77.0% correctly identified lack of iron in the diet as the main cause, compared to 66.2% in the control group, reflecting a notable positive knowledge gain associated with project exposure. Similarly, perceptions regarding vitamin A deficiency show improvement in the treatment group, where a higher share of respondents (34.3%) recognized it as a serious health concern compared to the control group (29.0%), alongside a lower proportion reporting uncertainty. These findings suggest that while basic dietary practices are already well established, the project has contributed more meaningfully to improving nutrition knowledge and awareness, particularly regarding micronutrient deficiencies. The remaining knowledge gaps—especially around identification of iodized salt and micronutrient risks—highlight the need for continued and more targeted nutrition education interventions.

A composite Food Hygiene Behavior Index (FHBI) was constructed using three indicators: handwashing before food handling, checking expiration dates before food preparation, and correct perception of food safety related to washing and peeling fruits and vegetables. Each indicator was scored as one if the recommended practice was followed, resulting in an index ranging from 0 to 3.

Table 13 Summary Index of Food Hygiene Behavior	
Group	Average FHBI
Treatment	2.07
Control	1.89

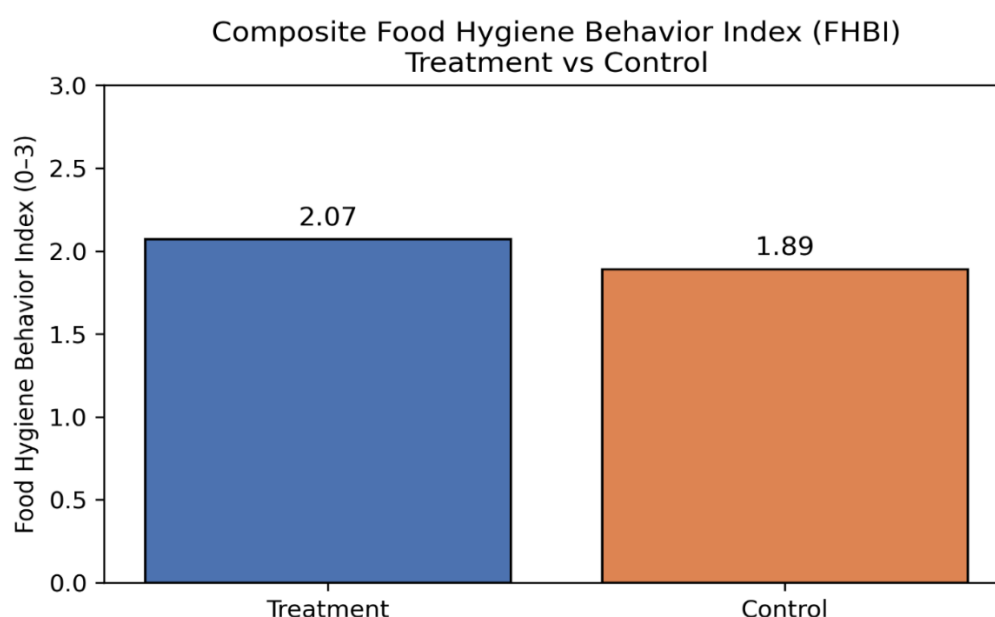


Figure 12 composite food hygiene behavior index

The results show that treatment households have a higher average FHBI (2.07) compared to control households (1.89) (Figure 12) indicating better overall food hygiene behavior among project beneficiaries. While hygiene practices are generally strong in both groups, the higher index score among treatment households reflects more consistent handwashing practices and better food safety

awareness. This suggests that project-supported nutrition and awareness activities have contributed to modest but meaningful improvements in household food hygiene behavior.

The results demonstrate clear and measurable progress against both the MDD-W and KAP outcome indicators. Women in treatment households report a higher average Dietary Diversity Score (DDS) of 3.96, compared to 3.51 among control households, indicating more diverse diets and improved nutritional adequacy among ETI-GB beneficiaries. This difference is driven by higher consumption of animal-source foods and pulses among treatment households, providing supportive evidence of progress against CI 1.2.8 (MDD-W). Improvements are also observed in household-level nutrition knowledge and practices, consistent with CI 1.2.9 (KAP). Awareness of micronutrient deficiencies is stronger among treatment households, with 77.0 percent correctly identifying lack of iron as the cause of anemia, compared to 66.2 percent among control households. Perceived seriousness of vitamin A deficiency is also higher in the treatment group (34.3 percent versus 29.0 percent), alongside reduced uncertainty. In terms of behavior, treatment households exhibit better food hygiene practices, reflected in a higher Food Hygiene Behavior Index (FHBI) score of 2.07, compared to 1.89 among control households.

These findings confirm that ETI-GB's nutrition-focused interventions translated into measurable improvements in women's dietary diversity and household nutrition-related knowledge and practices. While some gains remain modest and knowledge gaps persist—particularly regarding iodized salt identification—the consistent treatment–control advantages provide strong evidence that ETI-GB achieved the intended outcomes under CI 1.2.8 and CI 1.2.9, reinforcing the project's nutrition-sensitive impact pathways

5. Agricultural Production and Productivity Outcomes

This section assesses ETI-GB's contribution to improved agricultural production and productivity explicitly against the IFAD Core Outcome Indicators (COI) related to land access, adoption of improved practices, and production outcomes. In particular, the analysis examines performance against output indicators CI 1.1.1 (beneficiaries gaining increased secure access to land), CI 1.1.3 (rural producers accessing production inputs and technological packages), and CI 1.1.4 (persons trained in production practices and technologies), and their corresponding outcome indicators CI 1.2.1 (households reporting improved access to land for production), CI 1.2.2 (households reporting adoption of new or improved inputs, technologies or practices), and CI 1.2.4 (households reporting an increase in production). Using treatment–control comparisons of irrigated land expansion, crop-level production and commercialization data, and beneficiary perceptions, the section evaluates whether ETI-GB interventions translated output-level support into outcome-level improvements in production capacity, productivity, and adoption of improved agricultural practices. Findings are synthesized across irrigation and land development, value chain support, and farmer training interventions to reflect outcome-level changes rather than individual component performance.

Change in access to land

Table 14 shows a substantial difference in access to agricultural land between ETI-GB beneficiary (treatment) households and non-beneficiary (control) households. Overall, 89.2% of treatment households reported owning or having access to agricultural land, compared to 70.2% of control households, indicating a markedly higher level of land access among project beneficiaries. Disaggregating by land type reveals important structural differences. Nearly half of treatment households (48.4%) reported access to newly developed farmland, while no control households reported access to this category. This strongly suggests that irrigation and land development (ILD)

interventions under ETI-GB have been effective in bringing previously uncultivated land into productive use, thereby expanding the land base of beneficiary households. In contrast, access to old farmland is more prevalent among control households (86.4%) than treatment households (56.6%), reflecting the reliance of non-beneficiaries on traditionally cultivated land. This pattern indicates that ETI-GB beneficiaries are increasingly supplementing inherited or long-established farmland with newly developed plots, rather than depending solely on existing landholdings. Access to home gardens is more common among control households (14.7%) than treatment households (8.8%), suggesting that non-beneficiaries may rely more on small-scale, household-level cultivation, while beneficiaries are shifting toward larger or more commercially oriented plots enabled by project investments. The use of rented land is minimal in both groups (around 1%), indicating that land rental markets play a very limited role in agricultural production in the study areas. Similarly, access to shared land and communal pasture or grassland remains low across both groups, although control households report slightly higher reliance on communal resources, which may reflect more constrained private land access.

The findings highlight that ETI-GB has contributed to improved land access and structural transformation in agriculture, primarily through the expansion of newly developed farmland rather than through land rental or redistribution. This enhanced access to cultivable land provides a critical foundation for subsequent gains in agricultural production, crop diversification, and household income among beneficiary households.

Table 14 Ownership or Access to Agricultural Land		
Land Access / Type	Treatment Group (%)	Control Group (%)
Overall access to agricultural land	89.22	70.23
By land type		
Home garden	8.79	14.66
Own farmland (newly developed)	48.35	0.00
Own farmland (old)	56.59	86.38
Rented land	1.00	1.03
Shared land	2.75	1.38
Communal pasture / grassland	0.55	2.59
Other land arrangements	1.00	1.21

Table 15 presents a clear picture of how the Irrigation and Land Development (ILD) intervention has expanded productive landholdings among beneficiary households. At the baseline, the average irrigated landholding per household was 0.60 acres, reflecting the severe land and water constraints typical of high-mountain farming systems in Gilgit-Baltistan. Through the ILD intervention, households gained an additional 0.9 acres of irrigated land, which represents a substantial expansion of cultivable area rather than a marginal improvement. As a result, the total irrigated land after ILD increased to approximately 1.5 acres per household. In proportional terms, this indicates that ILD interventions more than doubled irrigated land availability for beneficiary households. Such an increase has important implications for agricultural productivity and livelihoods. The expansion of irrigated land enables households to bring previously barren or rain-fed plots under cultivation, diversify crops, increase cropping intensity, and reduce production risks associated with water scarcity.

Table 15 Irrigated Land Before /After ILD Intervention & Legal Status of newly Developed Land

Area	Irrigated Land Before ILD (acres) Baseline value	Additional Irrigated Land through ILD (acres)	Total Irrigated Land After ILD (acres)
Whole sample	0.60	0.90	1.50

FGDs provide strong and consistent evidence that ETI-GB interventions—particularly irrigation development and access infrastructure—enabled households to bring previously barren, abandoned, or underutilized land into productive use. In several locations, participants emphasized that land scarcity was not the primary constraint; rather, the absence of irrigation water and physical access had rendered land unusable. Once water channels were constructed and access improved, large areas of land were reclaimed for cultivation. In Dahimal (Ghizer) and Gohar Abad (Diamer), participants described transformative changes, noting that land which had remained uncultivated for decades was converted into productive agricultural plots. Similarly, in Surmo (Ghanche), participants reported that only a small fraction of households were cultivating land prior to the project, whereas after irrigation and road construction, the vast majority of households resumed farming activities. One participant explained that land which had been effectively abandoned due to distance and water scarcity was now actively cultivated by nearly all households.

“Before the project, most of the land in Broq was unused because it was too difficult to reach and there was no water. Now almost every household is cultivating and our agriculture land increased.”(FGD participant, Surmo, Ghanche).

These narratives directly reinforce the quantitative findings on expanded irrigated land and increased access to productive land, demonstrating that land development under ETI-GB resulted in genuine expansion of the agricultural land base rather than marginal adjustments.

Furthermore, this finding is strongly corroborated by KIIs with the Planning & Development Department and the Department of Irrigation & Water Management, who emphasized that ILD interventions addressed a binding structural constraint in Gilgit-Baltistan, where less than 2% of land is cultivable, by converting barren land into productive agricultural zones through technically sound and climate-resilient irrigation schemes

Table 16 Crop-wise Production, Sales, and Income of Treatment and Control Households					
Crop	Group	Harvested (in kg)	Sold (in kg)	% of Production Sold	Income (PKR)
Wheat	Treatment	565.80	12.52	2.20	1,716.52
	Control	415.82	16.95	4.10	2,515.72
Potatoes	Treatment	755.48	696.09	92.10	42,960.94
	Control	488.62	281.25	57.60	20,634.25
Vegetables	Treatment	137.38	48.44	35.30	12,819.67
	Control	153.67	60.00	39.00	13,098.62
Maize	Treatment	614.34	9.44	1.50	1,445.28
	Control	376.01	15.90	4.20	1,920.73
Alfalfa	Treatment	1,367.50	1,084.44	79.30	60,722.22
	Control	486.74	150.46	30.90	5,179.49
Apricot	Treatment	160.23	119.62	74.70	65,461.54
	Control	264.17	115.91	43.90	23,700.10
Apple	Treatment	144.75	102.33	70.70	37,083.33
	Control	181.16	66.08	36.50	19,903.23
Cherry	Treatment	304.20	284.00	93.40	123,000.00
	Control	267.23	199.59	74.70	85,825.40

The results highlight (Table 16) clear differences in production commercialization between treatment and control households. For staple crops such as wheat and maize, only a very small share of output is marketed in both groups, indicating that these crops remain largely subsistence oriented. In contrast, treatment households demonstrate substantially higher levels of commercialization for high-value crops. More than 90 percent of potato and cherry production, around 80 percent of fodder (alfalfa), and approximately 70–75 percent of fruit production (apricot and apple) are sold by treatment households, compared to markedly lower shares among control households. These commercialization gains are reflected in significantly higher crop incomes for treatment households, particularly for potatoes, fodder, and fruits, despite comparable or lower production volumes in some cases. Control households, while cultivating similar crops, tend to sell a smaller proportion of output and generate lower returns.

According to KIIs with Agriculture Extension and AKRSP, these productivity gains are closely linked to the adoption of improved seeds, vertical farming techniques, orchard rehabilitation, and year-round vegetable production made possible by expanded irrigation and targeted extension support. Overall, the findings suggest that ETI interventions have supported a shift from subsistence farming toward more market-oriented and higher-value agricultural production.

Figure 13 compares the share of households reporting increases in agricultural production between ILD beneficiaries (treatment group) and non-beneficiaries (control group). The results show a substantial and consistent advantage for treatment households. Nearly 61% of ILD beneficiary households reported an increase in agricultural production, compared to only 27% of control households. This large gap between treatment and control groups provides strong evidence that ILD interventions have played a decisive role in improving both production outcomes, beyond what could be attributed to broader trends or external factors affecting all households in the study areas.

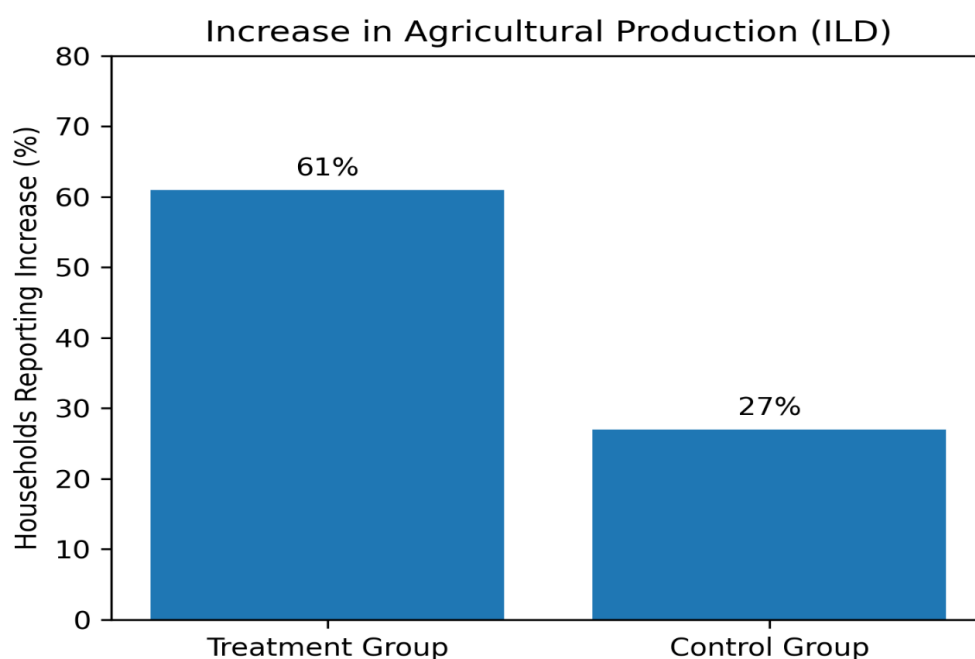


Figure 13 Increase in agricultural production

Table 17 examines the size of production gains among households that reported an increase. The distribution of gains differs markedly between treatment and control groups. Among ILD beneficiaries, a majority experienced moderate to large improvements: 44% reported significant production increases (50–99%), and 10% reported very large increases (doubling or more). In contrast, control households are more concentrated in the modest or very small increase categories, with 17% reporting very small production gains and only 1% experiencing very large increases.

Across all FGDs, participants consistently reported substantial increases in agricultural production following ETI-GB interventions. Increased production was observed across a range of crops, including cereals (wheat, maize), vegetables (tomatoes, potatoes, beans, chilies), fruits (apricots, apples), and fodder. In many cases, participants emphasized a shift from subsistence-oriented production toward surplus generation, with households producing beyond their own consumption needs for the first time. In Fina (Astora), improved farm-to-market roads were repeatedly cited as a key factor enabling higher production by reducing isolation, facilitating transport of inputs, and lowering post-harvest losses. Participants described agriculture transforming into a viable income-generating activity once surplus could reliably reach markets.

“Earlier, we grew crops only for ourselves because it was too difficult to take them to market. After the road, we can sell regularly, and production has increased many times.” (FGD participant, Fina, Astora)

Similarly, in Nagarkhas and Passu (Hunza), participants highlighted that vegetables which were previously difficult or impossible to grow, such as tomatoes and capsicum, were now produced consistently and in larger quantities, allowing households to both meet consumption needs and sell surplus

Table 17 Magnitude of Increase in Production (Among Households Reporting an Increase)		
Size of Increase	Production Increase	
	Treatment (%)	Control (%)
Very small (<20%)	4.80	17.09
Modest (20–49%)	40.80	50.63
Significant (50–99%)	44.00	31.01
Very large (≥100% / doubled)	10.40	1.27

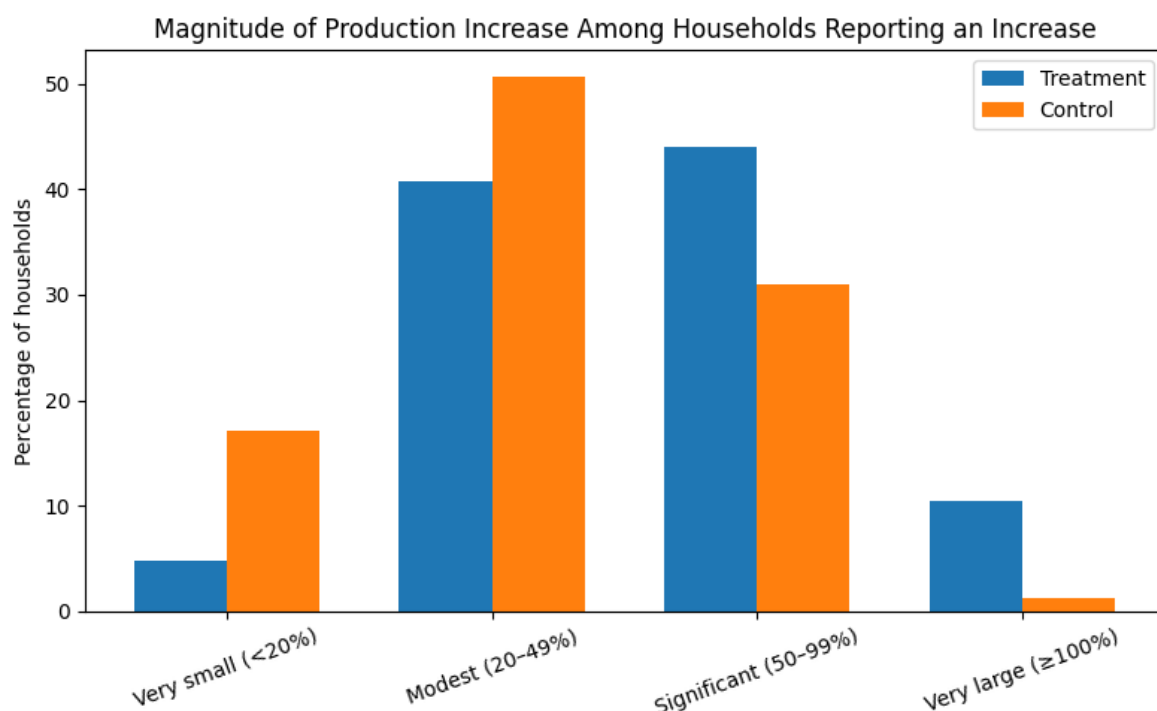


Figure 14 Magnitude of production increase

Table 18 identifies the key drivers behind observed increases in production. The expansion of irrigated area emerges as the dominant factor, especially among treatment households. Nearly 78% of ILD beneficiaries attribute production increases to expanded irrigation, compared to 58% of households controlled. The treatment households also report gains from fertilizer use and improved seed, though these factors are also cited by control households, suggesting that input use alone does not fully explain observed differences. Control households, by contrast, rely more heavily on general productivity factors and show lower reliance on market-related drivers such as improved prices or bargaining power. This suggests that ILD interventions strengthen both production capacity and market engagement, rather than focusing solely on input intensification.

Table 18 Key Factors Contributing to Increase in Agricultural Production		
Contributing Factor	Treatment (%)	Control (%)
Increase in irrigated area	78.40	58.23
Better yields due to improved seed	43.20	49.37
Better yields due to improved practices	32.80	42.41
Better yields due to fertilizer use	54.40	58.86
Other factors	5.60	3.80

The results indicate (Table 19 and Figure 15) that Value Chain (VC) support has had a substantial and meaningful impact on agricultural production. Overall, 86.27% of households reported an increase in the volume harvested following VC interventions, demonstrating the strong effectiveness of the support package in enhancing farm-level productivity.

Table 19 Magnitude of Production Change Attributed to VC Support	
Indicator	Total (%)
VC support increased production (volume harvested)	86.27
Among households reporting increase:	
Very small (<20%)	2.42
Modest (20–49%)	24.91
Significant (50–99%)	56.06
Very large (≥100%) / doubled	16.61

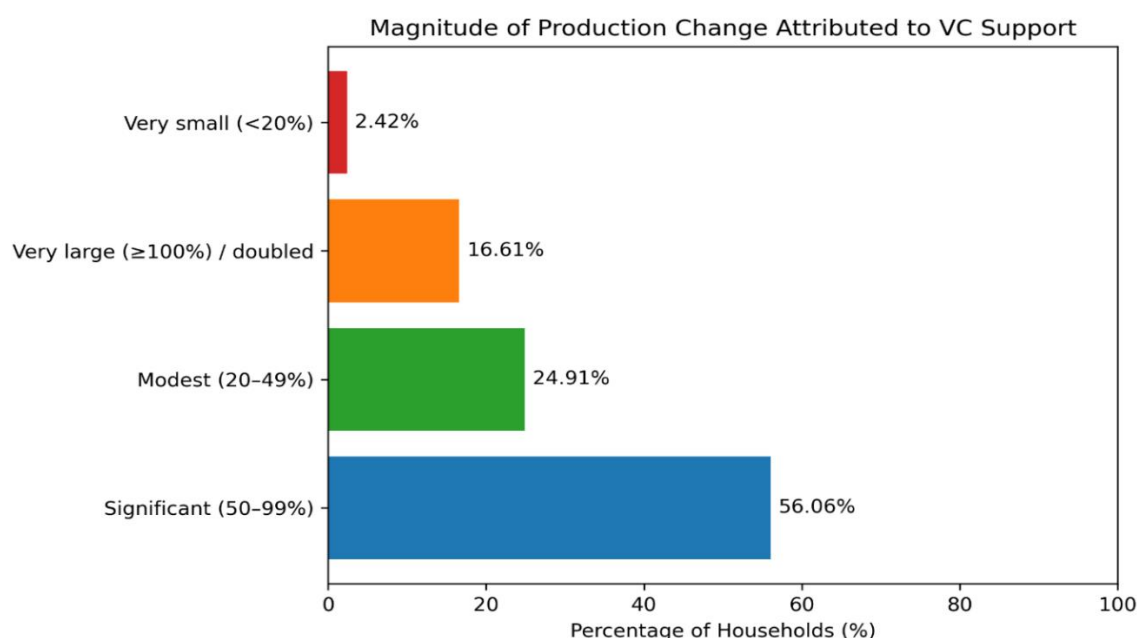


Figure 15 Magnitude of production change attributed to VC support

Among households experiencing an increase, the magnitude of change is largely moderate to high. A clear majority (56.06%) reported a significant increase in production (50–99%), while a further 16.61% achieved a very large increase (≥100%), effectively doubling production. In contrast, only a small fraction of households reported modest gains (24.91%) or very small improvements (2.42%). This distribution suggests that VC interventions are not only reaching a large proportion of beneficiary households but are also generating transformative productivity gains, rather than marginal improvements. The dominance of significant and very large increases is consistent with the combined effects of improved inputs, farmer training, and better market linkages under the VC component, reinforcing its role as a key driver of agricultural production growth within the project’s causal pathway.

Table 20 Factors Contributing to Increase in Agricultural Production in VC supported interventions	
Factor	Total (%)
Improved seeds	67.82
Fertilizer use	63.32
Irrigation	34.95
Farmer training	77.85
Mechanization / tools	48.79
Extension services	33.56

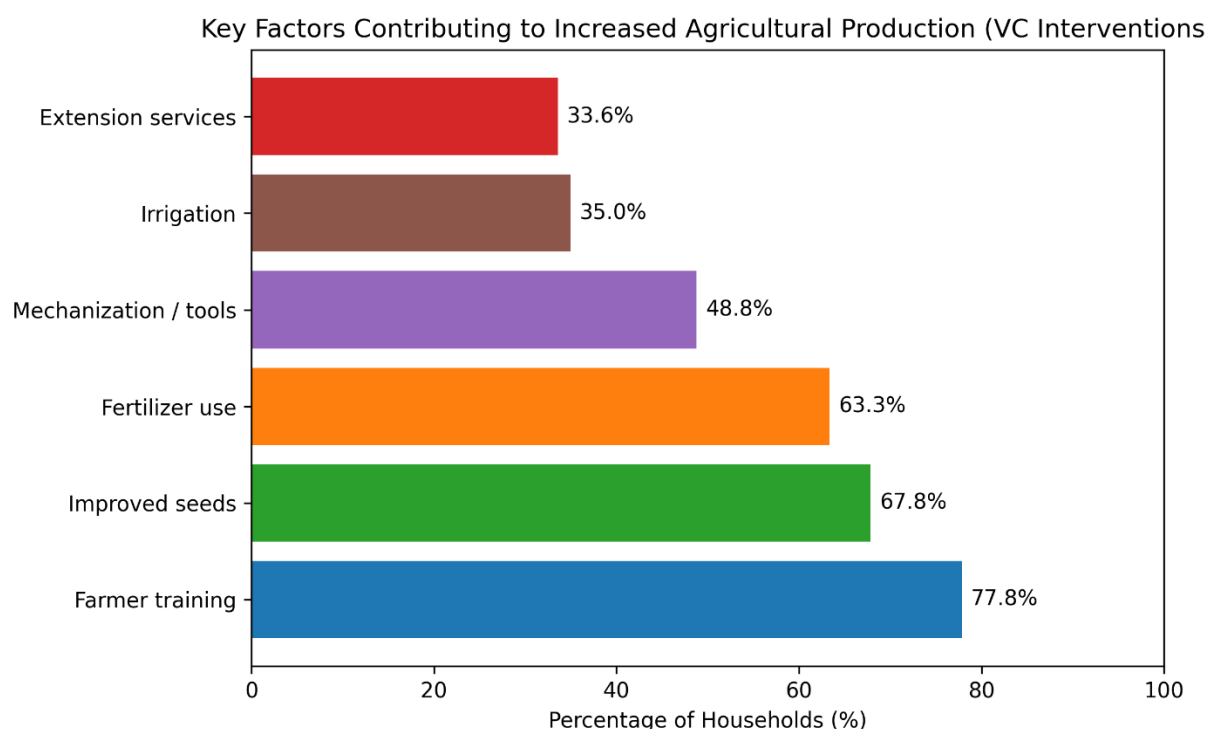


Figure 16 Key Factors Contributing to increased agricultural production under VC-support

The analysis of key factor contribution to increased agricultural production under VC-support (Table 20 and Figure 16) identifies farmer training as the most influential factor contributing to increased agricultural production under VC-supported interventions, cited by 77.85% of households (Table 26). This underscores the central role of capacity building in enabling farmers to effectively adopt improved practices and optimize input use. Improved seeds (67.82%) and fertilizer use (63.32%) also emerge as major drivers of production gains, reflecting the importance of timely access to quality inputs within the VC package. Secondary but still significant contributions come from mechanization and tools (48.79%), which help reduce labor constraints and improve operational efficiency. In contrast, irrigation (34.95%) and extension services (33.56%) are reported less frequently, suggesting either limited coverage or context-specific constraints in their effectiveness. Overall, the findings indicate that VC interventions generate production gains primarily through a combined pathway of skills development and improved input access, with complementary support from mechanization and infrastructure where available aligning well with the project's IFAD COI causal framework.

Table 21 Improved practices and access to agricultural inputs - VC	
	Overall (%)
Adoption of improved practices / technologies	87.76
Access to improved seeds / new varieties	80.60
Access to fertilizers / other inputs	81.19
Crop diversification	80.60

Table 21 shows strong adoption and access outcomes under the Value Chain (VC) interventions. Nearly 88% of beneficiary households adopted improved agricultural practices and technologies, indicating widespread behavioral change and uptake of promoted methods. Access to key inputs is also high, with 81.19% reporting improved access to fertilizers and 80.6% accessing improved seeds or new varieties. Similarly, 80.6% of households reported crop diversification, reflecting a shift toward more resilient and market-oriented production systems. Collectively, these results demonstrate that VC support has effectively translated into broad-based adoption of improved practices and enhanced input access, which are critical intermediate outcomes in the project's causal pathway. The high and relatively even levels across inputs and diversification suggest a well-integrated VC package, reinforcing productivity gains and supporting sustained income growth among beneficiary households.

FGDs reveal that productivity gains were achieved not only through land expansion but also, and often more importantly, through intensification and improved management of existing land. Participants across sites described adopting improved agronomic practices, quality seeds, better pest management, tunnel farming, mechanization, and improved post-harvest handling—all of which increased output per unit of land and labor. In Danyor and Passu, tunnel farming emerged as a particularly important driver of productivity. Participants emphasized that tunnels protected crops from pests and weather, stabilized yields, and enabled higher production from small plots of land.

“With tunnel farming, we get stable production even from small land. Earlier, crops were damaged, but now yields are much higher.” (FGD participant, Danyor)

In Churka (Shigar), Kuwardo (Skardu), and Singy (Kharmang), productivity gains were strongly linked to improved orchard management and post-harvest processing, especially for apricots. Participants described how training on spacing, harvesting, drying, and grading significantly reduced losses and increased effective output from the same quantity of raw produce.

“Before, most of our apricots were wasted. Now, with proper drying and sorting, we earn almost double from the same harvest.” (FGD participant, Churka, Shigar)

These accounts demonstrate that productivity improvements under ETI-GB were structural and sustained, driven by better knowledge, inputs, and practices rather than short-term increases

Table 22 Perceived Usefulness of Practices, Inputs, and Technologies	
Perception Category	(%)
Very useful	34.00
Useful	62.00
Neutral	4.00
Not useful at all	1.00
Total	100

Table 22 reports the results that show a strongly positive perception of the practices, inputs, and technologies promoted under the project. A clear majority of households (62%) rated them as useful, while a further 34% considered them very useful, indicating high perceived value and relevance. Only a small share of respondents reported neutral views (4%) or found the interventions not useful (1%). This overwhelmingly positive perception reinforces the high adoption and access rates observed under the VC interventions and suggests a strong likelihood of continued use and sustainability of improved practices. From an IFAD COI perspective, these findings confirm that the interventions are not only technically effective but also well accepted by beneficiaries, strengthening the causal pathway from awareness and access to adoption, productivity gains, and income improvements.

The findings provide clear, quantitative and qualitative evidence that ETI-GB achieved the intended results related to agricultural production and productivity. In line with the objective that beneficiaries gain increased and secure access to land (CI 1.1.1) and that households report improved access to land for production purposes (CI 1.2.1), access to farmland is substantially higher among beneficiary households. Specifically, 89.2 percent of treatment households reported access to agricultural land compared to 70.2 percent of control households, reflecting the contribution of irrigation and land development interventions. This is further supported by the expansion of irrigated land, with average irrigated holdings among ILD beneficiaries increasing from 0.60 acres at baseline to 1.5, strengthening productive capacity. Strong progress is also evident against the indicators related to technology uptake and skills development. Consistent with the objectives that rural producers access production inputs and technological packages (CI 1.1.3) and that persons are trained in production practices and technologies (CI 1.1.4), adoption of improved inputs and practices is widespread among beneficiaries. Nearly 88 percent of treatment households reported adopting at least one improved agricultural practice or technology, compared to significantly lower adoption among control households. Access to improved seeds and fertilizers exceeded 80 percent among beneficiary households, while 77.9 percent reported having received relevant agricultural training. These results directly translate into achievement of the outcome that households report adoption of new or improved inputs, technologies, or practices (CI 1.2.2).

Most importantly, these gains in land access and technology adoption resulted in measurable increases in agricultural production, fulfilling the outcome that households report an increase in production (CI 1.2.4). Among ILD beneficiary households, 61 percent reported increased agricultural production compared to 27 percent of control households. Moreover, the magnitude of change is substantial: 54 percent of treatment households experiencing increases reported production gains of 50 percent or more, including around 10 percent reporting that production had doubled. Similar patterns are observed among value-chain-supported households, where over 86 percent reported increased production volumes, and more than 70 percent experienced significant or very large gains. Taken together, these statistics demonstrate that ETI-GB effectively translated output-level investments in land development, input access, and farmer training into outcome-level improvements in production capacity and productivity. The consistent treatment–control differentials and the scale of reported production gains provide strong evidence that the project achieved its intended results under CI 1.1.1, CI 1.1.3, CI 1.1.4, CI 1.2.1, CI 1.2.2, and CI 1.2.4, fully aligning implementation evidence with the project’s results framework.

6. Market Access and Commercialization Outcomes

This section assesses the extent to which ETI-GB interventions improved market access and commercialization outcomes for beneficiary households and producer organizations, with particular reference to the outcomes that households report improved physical access to markets, processing, and storage facilities (CI 2.2.6) and that rural producers’ organizations report an increase in sales and

commercial performance (CI 2.2.5). The analysis examines whether investments in farm-to-market roads (FMRs), alongside complementary value-chain and organizational support, translated into reduced travel time, lower transaction costs, diversified marketing channels, increased volumes sold, and improved profitability of producer organizations. Drawing on household survey data, treatment–control comparisons, enterprise-level profit data, and qualitative evidence from FGDs, the section focuses on outcome-level changes in market participation and commercialization behavior, rather than on infrastructure delivery alone. Findings are synthesized to explain how improved physical connectivity and market linkages enabled households to shift from subsistence-oriented production toward market-oriented agriculture and supported producer organizations in capturing higher value from agricultural activities, consistent with the project’s intended impact pathways.

This study reveals that households report an average time saving of 3.5 hours per trip, reflecting a substantial reduction in travel time for accessing markets, services, and input suppliers (Table 23). Such a large time saving is particularly significant in remote and mountainous contexts, where long travel times previously constrained market participation and service access. Across all FGDs, participants consistently emphasized that FMRs dramatically reduced travel time, physical burden, and transport costs, enabling farmers to reach markets that were previously inaccessible. In Dahimal, Surmo, Fina (Astore), and Ghanche, respondents reported reductions from hours or even days of travel to 15–30 minutes, often replacing donkey or manual transport with tractors and vehicles. This qualitative evidence explains the high quantitative share of households using roads for market access.

Table 23 Time Saved Due to FMR

Indicator	Time (in hours)
Average time saved (hours)	3.50

KIIs with P&DD, C&W, and AKRSP provide institutional confirmation of these outcomes, noting that FMRs were explicitly designed to overcome geographic isolation, reduce transaction costs for smallholders, and improve the economic viability of agricultural commercialization in remote valleys. However, KIIs consistently cautioned that sustaining these gains will depend on the institutionalization of operation and maintenance responsibilities, as no dedicated departmental budgets or staffing are currently in place for long-term maintenance of irrigation infrastructure

FGDs provide strong explanatory evidence for the higher levels of market participation and income reported by treatment households. Improved farm-to-market roads dramatically reduced travel time, transport costs, and physical drudgery, particularly for women and elderly farmers. Participants consistently described how improved road access enabled them to sell produce more frequently, transport larger volumes, and access markets that were previously unreachable. In Diamer, Ghanche, and Skardu, communities emphasized that road connectivity transformed agricultural production from a largely subsistence activity into a market-oriented enterprise. This aligns closely with household survey results showing increased volumes sold and higher income among treatment households.

A participant of FGD at Surmo Ghanche stated that

“Earlier, we have challenges to go to our Broq. After constructing the road, now it is easy to carry down the grass, crops and all other stuff by tractor and other vehicles. Earlier, we had to carry all these manually, but now we bring them by tractor. Before the road construction, it takes more than one hour to reach there and now easily reach there in fifteen minutes”.

Beyond income, participants also highlighted indirect welfare benefits—such as improved access to health facilities and schools—which help explain observed improvements in health and nutrition indicators.

“Now one person can manage farming, and children regularly attend school.” (FGD, Ghanche)

In a focus group discussion, female beneficiaries from Danal Fina, Diamer district, were asked about the impact of the new or upgraded rural road on marketing and transportation costs. They noted that

“Now we have easy access to the market, even buyers from the market visit us, so we are cashing [in] our crops and vegetables from home, eventually this reduced our transportation cost up to 75%. In addition, this improved the number of cash crops and enabled us to link our products to better market through well-developed road and appropriate loader vehicles. Now we spend more time with our family and friends”.

Marketing Method	Treatment Group (FMR) Overall (%)	Control Group Overall (%)	Difference (Treatment – Control) (% pts)
Through middlemen	40.38	22.61	17.77
Direct contracts with buyers	12.91	4.98	7.93
Family sells directly at market	16.21	6.13	10.08
Family sells directly at farm gate	10.71	2.68	8.03
Through farmers’ group / cooperative	3.30	0.00	3.30
Other marketing methods	5.77	7.28	–1.51

Households benefiting from Farm-to-Market Roads (FMRs) exhibit markedly stronger engagement with market channels compared to non-beneficiaries (Table 24). The largest gains are observed in sales through middlemen and direct sales at markets and farm gates, reflecting reduced transport constraints and improved physical access to markets. The appearance of cooperative-based marketing among FMR beneficiaries, absent in the control group, further indicates early institutional and collective responses enabled by improved connectivity.

Table 25 Perceived Impacts of FMR on Production, Income, and Market Access

Indicator	Agree (%)	Strongly Agree (%)	Disagree (%)	Strongly Disagree (%)
Shift from self-consumption to market sales	73.15	15.34	11.51	0.00
Able to produce more due to easier selling	69.86	18.08	11.78	0.27
Generate cash income from agriculture	67.40	18.63	13.70	0.27
Post-harvest losses reduced due to easier transport	71.23	17.53	11.23	—
Easier for middlemen to buy at farm	67.40	21.92	10.14	0.55
Able to start new crops	70.41	18.63	10.96	—
More marketing opportunities available	73.97	23.84	1.92	0.27
New income-generating activities developed	75.07	23.84	1.10	—
Easier access to agricultural extension services	78.90	14.79	6.30	—
Easier access to agricultural inputs	78.90	13.97	7.12	—

In Table 25, the overall results indicate a strongly positive perception of Farm-to-Market Road (FMR) intervention among beneficiary households. More than 70% of respondents agree or strongly agree that FMRs have facilitated a shift from self-consumption to market-oriented production, improved production volumes, reduced post-harvest losses, and expanded marketing opportunities. Particularly high agreement is observed for access to extension services and agricultural inputs (~79%) and the development of new income-generating activities (~75%), underscoring the role of improved connectivity in strengthening both production and market access. These findings are fully consistent with the core outcome indicators on enhanced market participation, income generation, and institutional service access.

The FMR Impact Index is computed as an unweighted average of ten perception-based indicators, each measured as the percentage of respondents reporting “Agree” or “Strongly Agree.” Equal weighting was applied to reflect balanced importance across production, income, and market-access dimensions.

$$FMR\ Impact\ Index = \sum \frac{Composite\ Indicator\ Score}{10}$$

$$FMR\ Impact\ Index = 91.29\%$$

The overall FMR Impact Index of 91.3% indicates a very strong and consistent positive perception of Farm-to-Market Road interventions among beneficiary households. On average, more than nine out of ten respondents report improvements across key dimensions of market access, agricultural production, income generation, and service connectivity. This high index value provides consolidated evidence that FMR investments are effectively addressing structural market constraints and delivering outcomes, particularly those related to smallholder commercialization, income diversification, and improved access to services.

Table 26 presents the average profit per producer organization across districts, providing a standardized measure of enterprise performance. The results reveal substantial variation in profitability among ETI-supported producer organizations in Gilgit-Baltistan. Kharmang records the highest average profit per organization (PKR 361,267), followed by Diamer (PKR 346,683) and Hunza (PKR 264,400). These districts demonstrate relatively stronger enterprise performance, suggesting more favorable cost structures, better market access, or engagement in higher-value economic activities. Moderate levels of profitability are observed in Ghanche (PKR 253,078), Shigar (PKR 166,642), Gilgit (PKR 144,874), and Ghizer (PKR 134,900), indicating stable but comparatively smaller profit margins. In contrast, producer organizations in Skardu (PKR 27,986), Nagar (PKR 27,101), and Astore (PKR 17,714) report very low average profits per organization, highlighting potential constraints related to scale, efficiency, or market integration.

District	Profit per Organization (PKR)
Kharmang	361,267
Diamer	346,683
Hunza	264,400
Ghanche	253,078
Shigar	166,642
Gilgit	144,874
Ghizer	134,900
Nagar	27,101

FGDs consistently show that improved market access encouraged households to produce with the intention to sell, rather than only for household consumption. In Nagar, Passu (Hunza), and Danyor, participants described a clear behavioral shift: once households were confident that produce could be sold reliably, they invested more effort and inputs into production. In Nagarkhas, participants noted that vegetables previously grown only in small quantities are now cultivated at scale, with regular surplus sold in nearby markets. In Passu, farmers emphasized that reliable prices and access enabled them to plan production cycles around market demand rather than household needs alone. This shift represents a qualitative marker of commercialization, where production decisions are increasingly market-driven.

FGDs strongly indicate that value-chain (VC) support—including training, improved inputs, post-harvest handling, and buyer linkages—played a critical role in deepening commercialization. In Kuwardo (Skardu) and Churka (Shigar), participants described how improved grading, drying, and packaging of apricots enabled access to higher-value markets and reduced rejection rates. In Kuwardo, cooperative members reported that collective auctions and buyer agreements increased the volume of produce marketed and reduced post-harvest losses. Participants also discussed forward planning for expansion of chili cultivation under formal MoUs with buyers, indicating growing confidence in market demand and institutional arrangements. These narratives demonstrate that commercialization gains were not limited to selling more produce, but also involved selling better-quality produce into more structured markets.

Another recurring theme is the reduction of post-harvest losses and transaction costs, which effectively increased marketed surplus. Participants across Shigar, Kharmang, and Skardu described how improved drying sheets, tarpaulins, and storage practices allowed households to preserve

produce and sell it later at better prices, rather than distress-selling immediately after harvest. Reduced transport costs and improved logistics also lowered barriers to market entry, especially for smallholders. FGDs suggest that these reductions were particularly important for women and smaller producers, who previously lacked the capacity to access distant markets independently.

FGDs reveal that improved market access also facilitated greater participation of women in commercialization, particularly in processing and collective sales. In Kharmang and Shigar, women were actively involved in apricot drying, grading, and packaging, with VACs enabling them to link their processed products to buyers. Participants noted that income from these activities was often controlled by women and reinvested in household needs, reinforcing inclusive commercialization outcomes.

Both the quantitative and qualitative evidences demonstrate that ETI-GB has achieved strong and measurable progress toward improved market access and commercialization outcomes. Consistent with the objective that households report improved physical access to markets, processing, and storage facilities (CI 2.2.6), beneficiary households report an average time saving of 3.5 hours per trip when accessing markets and services. Qualitative evidence from FGDs across Astore, Ghanche, Diamer, and Skardu corroborates these findings, with participants reporting reductions from several hours—or even days—of travel to 15–30 minutes, alongside sharp declines in transport costs and physical drudgery. These improvements significantly reduced structural barriers to market participation in remote and mountainous contexts.

Improved physical connectivity translated into clear commercialization gains, supporting the outcome that rural producers' organizations report increased sales and market engagement (CI 2.2.5). Compared to control households, FMR beneficiaries demonstrate substantially higher engagement across multiple marketing channels, including sales through middlemen (40.4% vs. 22.6%), direct sales at markets (16.2% vs. 6.1%), farm-gate sales (10.7% vs. 2.7%), and direct contracts with buyers (12.9% vs. 5.0%). The emergence of cooperative-based marketing among beneficiary households—absent in the control group—signals early institutional responses enabled by improved connectivity and collective action.

Perception-based indicators further reinforce these outcomes. More than 70 percent of beneficiary households agree or strongly agree that FMRs facilitated a shift from self-consumption to market sales, enabled higher production due to easier selling, reduced post-harvest losses, and expanded marketing opportunities. Particularly high agreement is observed for improved access to agricultural inputs and extension services (approximately 79 percent) and the development of new income-generating activities (approximately 75 percent). These results are summarized in the FMR Impact Index of 91.3 percent, indicating a consistently strong positive effect across production, income, and market-access dimensions.

Commercialization gains are also reflected in the financial performance of ETI-supported producer organizations. Average profits per organization are substantial in several districts, reaching PKR 361,267 in Kharmang, PKR 346,683 in Diamer, and PKR 264,400 in Hunza, suggesting effective market integration and value capture where connectivity and organizational capacity are stronger. Lower profitability observed in districts such as Astore and Nagar highlights remaining constraints related to scale, efficiency, or market depth, underscoring the importance of complementary support beyond infrastructure alone.

Taken together, the evidence confirms that ETI-GB's investments in rural connectivity and market linkages significantly reduced transaction costs, expanded market participation, and strengthened

commercialization outcomes for both households and producer organizations. These results demonstrate clear achievement of the intended outcomes under CI 2.2.6 and CI 2.2.5, and position improved market access as a critical enabling pathway through which ETI-GB generated broader income, enterprise, and livelihood impacts.

7. Employment, Enterprises, and Value Addition

This section assesses ETI-GB's contribution to employment creation and enterprise-based livelihood opportunities, with specific reference to the outcome that beneficiaries obtain new jobs or employment opportunities (CI 2.2.1). The analysis examines whether ETI-supported rural enterprises—through skills development, enterprise-focused training, and facilitation of market linkages—generated employment beyond the household, supported participation of women and youth, and achieved financial viability sufficient to sustain income-generating activities. Using enterprise-level data on business types, management and employment linkages outside the household, and financial performance, the section evaluates employment outcomes as an economic effect of enterprise development, rather than as isolated project activities. Findings are interpreted to assess the scale, inclusiveness, and sustainability of employment generated under ETI-GB in the context of remote and structurally constrained mountain economies.

Rural enterprise development is a core pillar of the Economic Transformation Initiative (ETI) in Gilgit-Baltistan, aimed at strengthening livelihoods, promoting market-oriented activities, and enhancing household income in remote and fragile mountain contexts. Building on the region's strong dependence on agriculture and related value chains, ETI supports rural enterprises through skills development, enterprise-focused training, and facilitation of market linkages to enable households to move beyond subsistence production. This section presents an assessment of ETI-supported rural enterprises, focusing on enterprise types, participation beyond the household, training outcomes, and financial performance. Together, these indicators provide insights into how rural enterprise support under ETI is contributing to income generation, local economic activity, and the longer-term sustainability of livelihoods in Gilgit-Baltistan.

Table 27 Main Types of Rural Businesses Supported

Type of Support	Main types of business (N)	(%)
Trade of Agricultural Products	8	62.00
Processing (Crop)	5	38.00
Total	13	100.00

Table 27 presents the distribution of rural enterprises supported under the ETI intervention. The results indicate a clear emphasis on trade-oriented agricultural businesses, which account for 62% (8 out of 13) of the supported enterprises. These businesses are primarily engaged in the buying, selling, and marketing of agricultural products, reflecting ETI's role in strengthening market linkages, value chain integration, and commercialization of rural production. In contrast, crop-processing enterprises represent 38% (5 out of 13) of the supported businesses. Although fewer in number, these enterprises are strategically important as they contribute to value addition at the local level, reduce post-harvest losses, and create opportunities for employment and income diversification within rural communities. The distribution suggests that ETI support has been more strongly oriented toward facilitating market access and trade, while also laying an important foundation for local agro-processing activities. This balance indicates a gradual transition from primary production toward higher-value economic activities, enhancing the sustainability and resilience of rural livelihoods.

Table 28 Managers and Employment Outside the Household		
Category	Household Engagement Outside HH (≥ 1)	(%)
Managers (Females)	4	30.80
Managers (Males)	8	61.50
Managers (Youth)	3	23.10
Employees outside HH (Females)	2	15.40
Employees outside HH (Males)	7	53.80
Employees outside HH (Youth)	4	30.80

The findings reported in Table 28 indicate that ETI-supported rural enterprises have generated spillover engagement beyond the household, particularly through management and employment linkages. Male participation remains dominant, with 61.5% of households reporting at least one male manager outside the household, compared to 30.8% for females and 23.1% for youth. Similarly, employment effects are more pronounced for adult males (53.8% of households), while engagement of females (15.4%) and youth (30.8%) remains comparatively limited. It is important to note that these figures represent the prevalence of participation across households rather than mutually exclusive shares, as individual households may engage multiple categories simultaneously (e.g., employing both males and youth). Consequently, percentages reflect the breadth of participation across different groups rather than summing up to a fixed total.

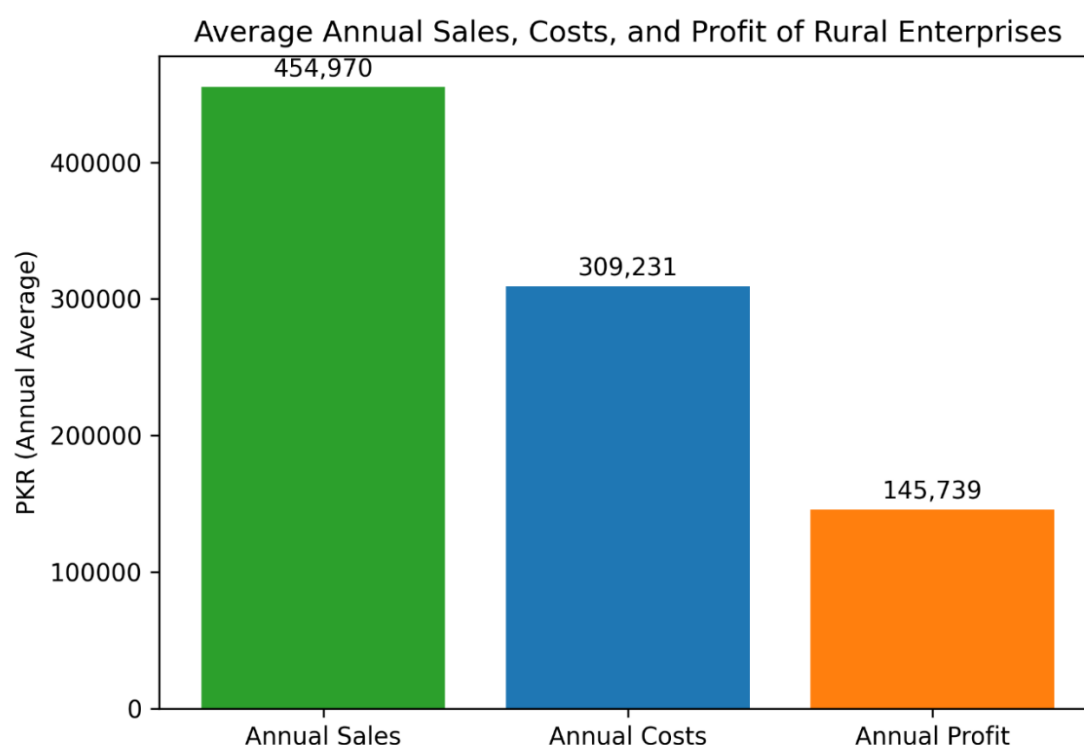


Figure 17 Average Annual Sales and Profit

The analysis of financial indicators shows (Figure 17) that ETI-supported rural enterprises generate substantial economic activity and positive returns. On average, enterprises report an annual value of sales of PKR 454,970, against annual operating costs of PKR 309,231, resulting in an average annual profit of PKR 145,739. This positive profit margin indicates that supported rural enterprises are, on

average, financially viable and capable of generating surplus income for beneficiary households. However, the relatively high variability in sales and costs across enterprises suggests significant differences in scale and performance, likely reflecting variations in enterprise type, market access, and managerial capacity. The presence of enterprises with very low or zero sales alongside high-performing enterprises highlights the need for continued technical support, market linkages, and business development services to improve consistency in outcomes.

The findings on rural enterprise development indicate that ETI has made meaningful progress in promoting market-oriented economic activities beyond primary agricultural production, particularly through trade-based enterprises and selected crop-processing initiatives. The dominance of trade-oriented businesses reflects a pragmatic response to the structural realities of Gilgit-Baltistan, where limited infrastructure, small market sizes, and high transaction costs constrain large-scale processing. At the same time, the presence of processing enterprises—though fewer in number—signals an important step toward local value addition, income diversification, and reduced post-harvest losses, which are critical for strengthening resilience in mountain economies. However, the relatively small number of supported enterprises suggests that the overall scale of enterprise development remains limited, and the observed impacts should therefore be interpreted as demonstrative rather than transformational at the regional level.

The evidence on management and employment linkages beyond the household points to positive spillover effects, particularly in terms of male employment and management roles. Nevertheless, the lower levels of female and youth engagement highlight persistent inclusion gaps, suggesting that enterprise development has not yet fully translated into broad-based employment opportunities for these groups. While qualitative engagement may exist, the quantitative results imply that additional, targeted measures may be required to overcome social, mobility, and skills-related barriers that constrain women's and youth participation in enterprise management and employment.

Financial performance indicators show that ETI-supported enterprises are, on average, profitable, indicating economic viability and the potential for sustained income generation. However, the wide dispersion in sales and profit levels reveals uneven performance across enterprises, with some operating at very small scales. This heterogeneity underscores the importance of continued business development support, including market intelligence, financial management skills, and stronger buyer linkages, to enhance enterprise stability and reduce the risk of stagnation or failure. Overall, while rural enterprise support under ETI demonstrates promising outcomes in terms of profitability and market engagement, its long-term contribution to inclusive and sustainable livelihood transformation will depend on scaling up successful models, strengthening institutional support, and addressing observed gender and youth participation gaps.

The evidence indicates that ETI-GB has generated positive but modest employment outcomes through rural enterprise development, consistent with the indicator that beneficiaries obtain new jobs or employment opportunities (CI 2.2.1). The analysis draws on data from a sample of 13 ETI-supported rural enterprises, which were selected for detailed assessment and are therefore indicative rather than exhaustive of the programme's enterprise portfolio. These sampled enterprises, primarily trade-oriented (62 percent) and crop-processing activities (38 percent)—illustrate the programme's role in promoting market-oriented economic activity beyond primary agricultural production.

Employment spillover effects beyond the household are evident within the sampled enterprises, particularly for adult males. Among these enterprises, 61.5 percent of households reported at least one male manager outside the household, while 53.8 percent reported employing at least one male worker, indicating that enterprise activities created employment linkages beyond family labor.

Engagement of women and youth, while present, remains comparatively lower, with 30.8 percent of enterprises reporting female managers, 23.1 percent reporting youth managers, and 15.4 percent and 30.8 percent reporting female and youth employees, respectively. These patterns suggest that while ETI-supported enterprises contributed to employment generation, inclusion of women and youth in enterprise management and wage employment remains constrained. Financial performance data from the sampled enterprises reinforce the potential viability of employment-supporting enterprises. On average, the assessed enterprises generated annual sales of PKR 454,970, against operating costs of PKR 309,231, yielding an average annual profit of PKR 145,739. This positive profit margin indicates that ETI-supported enterprises can sustain income-generating activities and associated employment. However, variation in sales and profitability across the sample underscores the need for continued business development support to strengthen enterprise performance and scale employment impacts.

These findings confirm that ETI-GB's rural enterprise support contributed to new employment opportunities and income generation, thereby fulfilling the intent of CI 2.2.1, while also highlighting that observed employment effects reflect demonstration-level impacts based on sampled enterprises. Scaling and deepening these outcomes will require expansion of successful enterprise models, stronger market integration, and targeted measures to enhance participation of women and youth.

8.Producer Organizations and Institutional Outcomes

This section assesses the performance of ETI-GB in strengthening producer-led institutional mechanisms, with explicit reference to the project's institutional and market-related core outcome indicators. In particular, the analysis examines whether rural producers' organizations were supported as functional collective institutions (CI 2.1.3), whether supported rural producers became members of rural producers' organizations (CI 2.1.4), whether producer organizations engaged in formal partnerships, agreements, or contracts with public or private entities (CI 2.2.3), and whether rural producers' organizations reported an increase in sales (CI 2.2.5). Using data from a survey of 84 ETI-supported producer organizations and Village Agriculture Cooperatives (VACs), complemented by qualitative evidence from FGDs and KIs, the section evaluates institutional reach, inclusiveness, service delivery, market engagement, and economic performance. Findings are organized around institutional outcomes and commercialization effects, to assess how collective action mechanisms contributed to value-chain integration and income generation in line with ETI-GB's intended impact pathways.

Producer organizations supported under the Economic Transformation Initiative (ETI) constitute a key institutional mechanism for promoting collective action, improving market access, and strengthening value chain integration in Gilgit-Baltistan. These ETI-supported organizations are designed to aggregate small producers, facilitate access to services, and enhance the economic viability of rural livelihoods in remote and mountainous contexts. This section presents an assessment of ETI-supported producer organizations based on data collected through the household survey. The analysis draws on information covering membership and participation, leadership characteristics, main economic activities, employment generation, and the range of services provided by these organizations. It also examines the financial performance of producer organizations and their engagement in formal partnerships, agreements, or contracts with public and private entities.

This study sampled 84 producer organizations/VACs drawn from ten districts. The analysis of membership affiliation with Village/ Value Chain Associations across districts indicates strong overall participation, with a total of 27,951 members affiliated under ETI-supported structures (Table 29). At

the aggregate level, male members constitute the largest share (47.5%), followed by youth (26.9%) and female members (25.6%), suggesting a relatively balanced inclusion of youth alongside adult members, while women's participation, although substantial, remains comparatively lower. District-wise patterns reveal notable variation in the composition of VAC membership.

Diamer accounts for the largest number of affiliated members (6,528), with a strong male dominance (57.5%), while female participation remains limited (9.1%). In contrast, districts such as Gilgit and Hunza exhibit relatively higher levels of female inclusion, where women account for 48.0% and 44.3% of VAC members, respectively, indicating more gender-balanced participation in these districts. Youth participation is particularly prominent in Astore (39.2%), Diamer (33.4%), Shigar (31.0%), and Kharmang (30.0%), reflecting the effectiveness of VACs in engaging younger populations in value-chain-related activities. Districts such as Ghizer and Nagar display more evenly distributed participation across male, female, and youth categories, suggesting inclusive membership structures.

Table 29 Number of Members Affiliated with VAC

	Number of MALE Members affiliated with VAC		Number of FEMALE Members affiliated with VAC		Number of YOUTH Members affiliated with VAC		Total
District	Count	%	Count	%	Count	%	Count
Astore	971	34.92	720	25.89	1,090	39.19	2,781
Diamer	3,754	57.51	594	9.10	2,180	33.39	6,528
Ghanche	2,444	51.01	1,122	23.42	1,225	25.57	4,791
Ghizer	577	45.50	438	34.54	253	19.95	1,268
Gilgit	795	37.61	1,014	47.97	305	14.43	2,114
Hunza	1,080	45.32	1,056	44.31	247	10.37	2,383
Kharmang	764	38.24	634	31.73	600	30.03	1,998
Nagar	440	47.67	337	36.51	146	15.82	923
Shigar	1,014	43.48	595	25.51	723	31.00	2,332
Skardu	1,435	50.65	644	22.73	754	26.61	2,833
Grand Total	13,274	47.49	7,154	25.59	7,523	26.91	27,951

The district-wise analysis of total sales values indicates (Table 30) substantial variation in the scale of economic activity across ETI-supported producer organizations in Gilgit-Baltistan. Diamer stands out with the highest average annual sales (PKR 11.24 million) and a median of PKR 10.0 million, reflecting the presence of relatively larger or more commercially active organizations and stronger market integration in the district. A second tier of districts with comparatively high average sales includes Astore (PKR 3.14 million), Shigar (PKR 2.76 million), and Nagar (PKR 2.31 million). In these districts, the gap between average and median sales—particularly in Shigar, where the median (PKR 0.7 million) is substantially lower than the average—suggests the presence of a few high-performing organizations alongside many smaller-scale entities, indicating unequal distribution of sales performance. Moderate sales levels are observed in Ghizer (PKR 1.81 million), Hunza (PKR 1.82 million), Skardu (PKR 1.51 million), and Ghanche (PKR 1.27 million). In these districts, relatively lower median values compared to averages point to skewed sales distributions, where a limited number of organizations account for a disproportionate share of total sales. At the lower end of the spectrum, Gilgit (PKR 0.54 million) and Kharmang (PKR 0.62 million) report the lowest average sales, with medians close to or slightly above their averages, suggesting more uniform but low-volume commercial activity.

Table 30 Total Sales Value Last Year		
District	Average Sales (Million PKR)	Median (Million PKR)
Astore	3.14	2.25
Diamer	11.24	10.00
Ghanche	1.27	0.70
Ghizer	1.81	0.80
Gilgit	0.54	0.40
Hunza	1.82	0.96
Kharmang	0.62	0.66
Nagar	2.31	1.50
Shigar	2.76	0.70
Skardu	1.51	0.55

Table 31 presents the range of services offered by ETI-supported producer organizations, highlighting their multi-functional role in supporting rural livelihoods and value-chain development. Among the services, training emerges as the most frequently reported, accounting for 23.4% of all responses. This underscores the strong emphasis placed by ETI on capacity building and skills development to enhance managerial, technical, and entrepreneurial capabilities of members. Closely following is access to market and marketing services (21.5%), reflecting the critical role of producer organizations in improving market linkages, aggregation, and commercialization of agricultural products. The provision of purchase of equipment or inputs (18.8%) and crop processing facilities (15.1%) further highlights efforts to support productivity enhancement and value addition at the local level. Credit and loan services (12.6%) indicate a growing role of producer organizations in facilitating access to finance, although this remains less prominent compared to training and market-related services. In contrast, storage facilities (7.1%) and other services (1.5%) are relatively limited, suggesting potential areas for further strengthening, particularly to reduce post-harvest losses and improve supply-chain efficiency.

Table 31 Main services provided by the cooperative/organization		
Description	N	(%)
Access to market and marketing	70	21.54
Credit/loan	41	12.62
Crop Processing facility	49	15.08
Other	5	1.54
Purchase of equipment or inputs	61	18.77
Storage Facility	23	7.08
Training	76	23.38
Grand Total	325	100.00

Out of the 84 ETI-supported producer organizations surveyed, 28 organizations (33.3%) reported being engaged in formal partnerships, agreements, or contracts with public or private entities (Table 32). This indicates that approximately one-third of the organizations have established formal institutional linkages, while the majority are still operating without structured partnerships. The extent of engagement in formal partnerships varies considerably across districts. Hunza shows the highest level of engagement, with 100% of surveyed organizations reporting formal partnerships, followed by Gilgit (85.7%) and Nagar (80.0%), suggesting stronger institutional connectivity and market integration in these districts. Ghizer also demonstrates a relatively high level of engagement (66.7%), while Skardu (40.0%) and Astore (28.6%) show moderate levels of partnership formation. In contrast, organizations in Diamer (11.8%) and Shigar (22.2%) report limited engagement in formal

partnerships, while Ghanche and Kharmang report no formal partnerships at all. These district-level disparities highlight differences in institutional capacity, access to external actors, and enabling environments for partnership development.

District	Count	%
Astore	2	28.57
Diamer	2	11.76
Ghanche	0	0.00
Ghizer	2	66.67
Gilgit	6	85.71
Hunza	6	100.00
Kharmang	0	0.00
Nagar	4	80.00
Shigar	2	22.22
Skardu	4	40.00
Grand Total	28	33.33

Qualitative insights were gathered through focused group discussions and key informant interviews with members and office bearers of producer organizations, including cooperative societies (VACs), across ETI intervention areas. These discussions explored institutional functioning, changes in farming practices, access to inputs and markets, income effects, women's participation, and sustainability mechanisms. KIIs consistently identified Village Agriculture cooperatives as key institutional mechanisms underpinning these outcomes, enabling farmers to pool produce, access inputs, receive training, and connect with markets

Across Shigar, Skardu, Kharmang, and Hunza, cooperative societies emerged as the first functional institutional platforms that moved communities beyond awareness-raising toward practical value chain engagement. Prior to ETI support, communities reported fragmented and informal practices, with no structured mechanisms for collective marketing, pricing, or quality control. ETI-supported cooperatives introduced organized production, collective aggregation, improved post-harvest handling, and buyer linkages, transforming subsistence-oriented farming into semi-commercial activity

“Before ETI, there were only awareness sessions, but no real value-chain work. ETI was the first project that brought practical benefits through the cooperative. Earlier nothing was structured; now we have systems for collection, grading, and selling.” (FGD participant, Cooperative Society – Shigar)

FGDs consistently highlighted that cooperatives facilitated access to improved seeds, saplings, drying sheets, tarpaulins, and technical training, resulting in measurable improvements in yield, quality, and prices. Apricot value chains show the strongest evidence: collective drying, sulfur treatment, grading, and auctioning through cooperatives increased prices from PKR 20–40/kg to PKR 235–400/kg. These gains were repeatedly linked to cooperative-led systems rather than individual effort.

“Earlier, no one would buy our apricots even for 20–25 rupees per kilo. After collective drying and marketing through the cooperative, we sold at 235 to 400 rupees per kilo. This change came only because of the cooperative system.” (FGD participant, Cooperative Society – Skardu)

“If you go to the market alone, you get a low price. When you go through the cooperative, buyers pay more because they trust the quality and the system.” (FGD participant, VAC/Cooperative – Passu, Hunza)

“Earlier, most apricots were wasted, but this year we collected 65,000 kg through the cooperative. Women earned between 20,000 to 80,000 rupees per household. This income helped with food, school fees, and farm inputs.” (FGD participant, Cooperative Society – Kharmang)

The analysis of ETI-supported producer organizations provides strong evidence that collective institutional mechanisms have played a pivotal role in enhancing market access, value-chain integration, and income generation in Gilgit-Baltistan. The large scale of membership of nearly 28,000 affiliated members across districts demonstrates the reach and institutional embeddedness of VACs as a platform for collective action. District-level variations in gender and youth participation highlight both progress and persistent inclusion gaps: while several districts exhibit relatively balanced participation, others—particularly Diamer—remain male-dominated, suggesting that social norms and local contexts continue to shape participation outcomes. These differences underscore the importance of tailoring institutional support to district-specific socio-cultural conditions rather than relying on a uniform model.

Economic performance indicators reveal that producer organizations have generated meaningful commercial activity, but with highly uneven outcomes across districts. The strong concentration of sales and profits in a limited number of districts and organizations indicates that while some producer organizations have successfully transitioned toward commercially viable operations, many continue to operate at small scales with limited market integration. This heterogeneity suggests that ETI-supported producer organizations are at different stages of institutional and economic maturity. High-performing districts appear to benefit from better market connectivity, stronger leadership, and more effective engagement with buyers, whereas low-performing districts face constraints related to scale, infrastructure, and access to reliable markets.

The service portfolio of producer organizations reflects a well-aligned intervention strategy, with a strong emphasis on training and market facilitation. However, the relatively limited provision of storage and processing services points to structural gaps that may constrain further value addition and increase vulnerability to post-harvest losses. Similarly, while one-third of organizations have established formal partnerships, the majority remain outside formal contractual arrangements, indicating that institutional linkages with public and private actors are still nascent. District-level disparities in partnership engagement further suggest that enabling environments and institutional capacity vary widely, affecting the ability of producer organizations to formalize market relationships.

Qualitative evidence strongly reinforces the quantitative findings by illustrating how cooperatives transformed fragmented and informal practices into organized systems of production, quality control, and collective marketing. The dramatic increases in apricot prices and documented income gains—particularly for women—highlight the transformative potential of cooperative-led value-chain systems. At the same time, these success stories also point to the risk of uneven impact, as benefits may concentrate among more active or better-organized cooperatives. Overall, the findings suggest that ETI-supported producer organizations have laid a solid foundation for collective market engagement and income growth, but sustaining and scaling these impacts will require continued institutional strengthening, deeper market partnerships, and targeted efforts to address geographic and gender-based disparities in participation and performance.

The evidence demonstrates that ETI-GB made substantial progress in supporting functional rural producers' organizations (CI 2.1.3) and expanding producer participation in collective institutional structures (CI 2.1.4). Across ten districts, 84 producer organizations/VACs were assessed, with a total membership of 27,951 affiliated producers, indicating strong institutional reach and embeddedness. Membership composition shows meaningful inclusion of youth (26.9%) alongside adult members, while women's participation (25.6%)—though substantial—remains uneven across districts, with strong inclusion in Hunza and Gilgit but persistent gender gaps in districts such as Diamer. Clear evidence is also observed against the indicator that rural producers' organizations engage in formal partnerships or contractual arrangements with public or private entities (CI 2.2.3). One-third of surveyed organizations (33.3%) reported active formal partnerships, with particularly strong engagement in Hunza (100%), Gilgit (85.7%), and Nagar (80.0%). At the same time, very low or absent partnership formation in districts such as Ghanche, Kharmang, and Diamer highlights uneven institutional maturity and enabling environments across the programme area.

Commercial performance indicators confirm progress toward the outcome that rural producers' organizations report an increase in sales (CI 2.2.5), although with significant heterogeneity. Average annual sales exceed PKR 11.2 million in Diamer and range between PKR 2–3 million in districts such as Astore, Shigar, and Nagar, while remaining below PKR 1 million in Gilgit and Kharmang. The divergence between average and median sales values in several districts indicates that a smaller number of high-performing organizations account for a disproportionate share of total sales, suggesting unequal commercialization outcomes across producer organizations. Service provision patterns further illustrate institutional functionality, with producer organizations primarily delivering training (23.4%), market access and marketing services (21.5%), and input or equipment support (18.8%), directly supporting members' production and commercialization capacity. However, relatively limited provision of storage facilities (7.1%) and processing services (15.1%) points to structural constraints that may limit further value addition and risk mitigation.

Qualitative evidence strongly reinforces these findings, demonstrating how ETI-supported cooperatives transformed fragmented and informal production systems into organized structures for collective aggregation, quality control, and marketing. Reported increases in apricot prices—from PKR 20–40/kg to PKR 235–400/kg—and associated income gains for women and households underscore the transformative potential of cooperative-led value-chain engagement, while also highlighting the risk of uneven benefits across districts and organizations.

In sum, the findings confirm that ETI-GB successfully strengthened producer organizations as collective economic institutions and enabled meaningful gains in market engagement and sales, thereby fulfilling the intent of CI 2.1.3, CI 2.1.4, CI 2.2.3, and CI 2.2.5. At the same time, the wide variation in inclusion, partnership formation, and commercial performance indicates that producer organizations are at different stages of institutional maturity, underscoring the need for continued capacity building, deeper market linkages, and targeted support to address geographic and gender-based disparities.

9. Access to Agricultural Finance and Sustainability: Partial Credit Guarantee Scheme (PCGS)

While ETI-GB significantly improved agricultural productivity and institutional organization through irrigation development, value chain support, and the establishment of Village Agriculture Cooperatives (VACs), beneficiary farmers continued to face structural barriers in accessing formal agricultural finance. High interest rates, collateral requirements, and complex banking procedures have historically excluded smallholders in Gilgit-Baltistan from commercial credit markets,

constraining enterprise expansion and the sustainability of income gains. The State Bank of Pakistan estimates the agricultural financing requirement of the region at over PKR 7.0 billion, underscoring the scale of the financing gap faced by farmers

To address this constraint and ensure sustainability beyond grant-based support, ETI-GB operationalized the Partial Credit Guarantee Scheme (PCGS) through a tripartite partnership with the Gilgit-Baltistan Rural Support Programme (GBRSP) and HBL Microfinance Bank Ltd. The scheme integrates VAC members into formal financial systems through tailored agricultural loan products, interest rate subsidies of up to 50 percent, simplified documentation requirements, and a partial risk-sharing mechanism on defaulted loans. VACs serve as the primary platforms for borrower identification, referral, and monitoring, thereby reinforcing institutional accountability while facilitating access to finance

As of 31 December 2025, PCGS enabled the disbursement of PKR 1.29 billion in agricultural credit to 4,232 VAC member farmers across Gilgit-Baltistan, with women accounting for 27 percent of borrowers and youth for 17 percent. By transitioning farmers from grant dependence to subsidized commercial credit, the Partial Credit Guarantee Scheme represents a critical sustainability pathway for ETI-GB, supporting continued investment in productive assets, enterprise growth, and income generation beyond the project lifecycle.

10. Agri-Tech Fellows Programme: Strengthening Human Capital and Innovation

Recognizing that long-term agricultural transformation requires sustained investments in human capital and knowledge systems, ETI-GB introduced the Agri-Tech Fellows Programme to mainstream agricultural education, innovation, and nutrition awareness within schools and farming communities. Implemented in partnership with the School Education Department, Government of Gilgit-Baltistan, and GBRSP, the programme addresses gaps in youth engagement, agricultural skills, and technology diffusion that constrain the sustainability of rural development outcomes

Through a competitive selection process, ETI-GB deployed 50 Agri-Tech Fellows across 82 government schools in 10 districts, reaching more than 8,000 students. The fellows delivered structured learning on agricultural entrepreneurship, value chain development, modern production technologies, kitchen gardening, nutrition, and climate-resilient practices. Each fellow was also linked to at least one nearby VAC, where they provided technical support to farmers and disseminated awareness messages on food security and nutrition, thereby strengthening linkages between formal education systems and community-based agricultural institutions. By bridging schools and VACs, the Agri-Tech Fellows Programme enhanced the depth of ETI-GB's institutional and capacity-building interventions, promoted intergenerational knowledge transfer, and reinforced the adoption of improved practices critical for sustaining project outcomes.

Chapter Four: Conclusion

This impact assessment examined the performance of the Programme against its logframe and the Core Outcome Indicators (COIs), using evidence collected at the time of this assessment through a mixed-methods, quasi-experimental approach. The findings provide strong and credible evidence that ETI-GB has generated substantial progress across most outcome and goal-level indicators, particularly those related to access to productive assets, agricultural production, employment generation, income growth, market access, institutional strengthening, and poverty reduction. The consistency of results across quantitative and qualitative data sources strengthens confidence in the attribution of observed outcomes to programme interventions.

A key conclusion emerging from the assessment is that ETI-GB has successfully transformed the productive asset base of beneficiary households. Investments in irrigation and land development significantly expanded access to cultivable land and reliable water supply, enabling households to bring previously unproductive land into use and reduce production risk. These asset-based interventions created the foundation upon which improvements in productivity, income, and resilience were built, demonstrating the central importance of secure access to land and water in the mountainous context of Gilgit-Baltistan.

The assessment further concludes that ETI-GB has achieved strong gains in agricultural productivity through a combination of improved asset access, technology adoption, and extension support. High levels of adoption of improved agricultural practices among beneficiary households translated into marked increases in agricultural production, substantially outperforming non-beneficiary households. These results confirm that productivity gains were not incidental but rather the outcome of deliberate programme design that integrated physical infrastructure with capacity building and technical support.

Employment generation and income growth represent another major area of success. In line with the logframe indicator on beneficiaries with new jobs or employment opportunities (CI 2.2.1), ETI-GB interventions expanded both on-farm and off-farm employment through increased agricultural activity, value chain engagement, cooperative operations, and rural enterprises. Income gains were widespread and exceeded programme targets, with most beneficiary households reporting substantial improvements. Importantly, these income gains were driven by increased production and employment rather than short-term or external factors, indicating a higher likelihood of sustainability.

The assessment also highlights the critical role of improved market access in enabling rural economic transformation. Investments in farm-to-market roads significantly reduced travel time and transaction costs, allowing households to engage more frequently and effectively with markets. Improved connectivity amplified the benefits of increased production and employment by facilitating higher marketed surplus, reducing post-harvest losses, and improving access to services. These findings underscore the importance of addressing spatial constraints in remote and mountainous regions.

Institutional strengthening through the establishment and support of Village Agriculture Cooperatives emerged as a key enabling factor for sustaining programme impacts. Producer organizations improved access to inputs, facilitated collective action, and supported enterprise development, contributing to improved profitability and market participation. However, while progress in institutional development has been substantial, the assessment finds that formal buyer linkages and contractual market arrangements remain limited, indicating an area where further effort is required to consolidate value chain outcomes.

Beyond economic outcomes, the assessment concludes that ETI-GB has generated positive spillover effects on nutrition, health, and overall household well-being. Improvements in dietary diversity, food hygiene practices, and access to healthcare reflect the broader welfare impacts associated with increased income stability, improved food availability, and enhanced physical access to services. These outcomes highlight the multidimensional nature of ETI-GB's impacts and its contribution to human development alongside economic transformation.

At goal level, the assessment finds strong evidence of poverty reduction among beneficiary households. The decline in poverty incidence between the baseline and the time of this assessment substantially exceeds the programme's target, indicating that ETI-GB has contributed meaningfully to improving living standards and reducing vulnerability. The evidence suggests that poverty reduction under ETI-GB is closely linked to durable changes in productive capacity and employment opportunities, rather than temporary coping strategies, thereby strengthening household resilience.

The assessment concludes that ETI-GB is on a strong trajectory toward achieving its intended development objectives. The integrated design of the programme—linking productive assets, markets, institutions, and human development outcomes—has proven effective in the challenging high-mountain context of Gilgit-Baltistan. While some areas, particularly deeper market integration and institutional maturation, warrant continued attention, the overall evidence provides a solid foundation for consolidating gains, informing adaptive management, and scaling the ETI-GB model in similar contexts.

1. Recommendations

The findings of this impact assessment indicate that ETI-GB has generated strong and largely attributable results across productive assets, agricultural production, employment, income, market access, institutional strengthening, and poverty reduction. The recommendations below are therefore framed to consolidate proven successes, address identified gaps, and enhance the sustainability and scale of observed impacts.

1. The assessment demonstrates that investments in irrigation and land development have been foundational to all subsequent livelihood and income gains. To sustain these gains, future programme efforts should place greater emphasis on the long-term operation and maintenance of irrigation infrastructure. Strengthening community-based water management arrangements and providing periodic technical backstopping would help protect productivity gains and ensure that expanded irrigated land continues to deliver benefits under changing climatic conditions.
2. Findings show that high adoption of improved agricultural practices was a key driver of increased production and income among beneficiary households. Building on this success, extension and advisory services should be further institutionalized to ensure continued learning beyond the project period. Refresher trainings, peer learning, and targeted support for women and smaller landholders would help sustain productivity gains and reduce the risk of reversion to traditional practices.
3. The assessment confirms that employment creation under ETI-GB has occurred primarily through expanded agricultural activity, value chain engagement, and cooperative operations. To strengthen these employment pathways, future interventions should more explicitly track employment outcomes and support diversification into higher-value activities such as processing, storage, grading, and packaging. These activities offer scope for additional employment generation, particularly for youth and women, and can deepen the economic impact of value chain interventions.

4. Improved market access emerged as a critical enabling factor for income growth and employment. While farm-to-market roads have significantly reduced transaction costs, the findings suggest that market integration remains largely informal for many producer organizations. Strengthening formal buyer linkages through contractual arrangements, quality assurance mechanisms, and market information systems would help consolidate gains and reduce exposure to price volatility. Support to producer organizations in meeting quality standards and consistent supply requirements would further enhance market integration.
5. Institutional strengthening through Village Agriculture Cooperatives has contributed positively to access to inputs, collective action, and enterprise profitability. However, the assessment also indicates variation in organizational maturity and market engagement across cooperatives. Continued investment in governance, financial management, and leadership capacity of producer organizations would help ensure their long-term viability. Where appropriate, encouraging federated or networked structures could enhance economies of scale and bargaining power.
6. The findings highlight positive spillover effects on nutrition, health, and household well-being, driven largely by increased income and food availability. To reinforce these outcomes, future programming should more explicitly integrate nutrition-sensitive approaches into agricultural and value chain interventions. Strengthening linkages with health and nutrition services, and continuing to target women as key agents of change, would help sustain and amplify welfare impacts.
7. Finally, the overall pattern of results suggests that the integrated ETI-GB model—linking productive assets, markets, institutions, and human development outcomes—is well suited to the high-mountain context of Gilgit-Baltistan. Future scale-up or replication efforts should retain this integrated design, while allowing for contextual adaptation. Documenting and disseminating lessons from ETI-GB would support learning within government and among development partners and inform the design of similar interventions in other remote and fragile regions

Annexures

Table Annex. A Progress Comparison Matrix over Time

COI Indicator	Quantitative Measure Used in Report	Baseline	2022	Treatment in 2025
CI 1.1.1 Beneficiaries gaining increased secure access to land (acres)	Avg. irrigated land	0.6	1.5	1.5
CI 1.2.1 HHs reporting improved access to land for production (%)	HHs	0	69	89.2
CI 1.1.3 VACs accessing inputs/tech packages (%)	HHs	0	28	69.5
CI 1.2.2 HHs adopting new/improved practices (%)	Adoption rate	0	66	78.5
CI 1.2.4 HHs reporting increase in production (%)	HHs	0	59	60.9
CI 1.1.8 HHs receiving nutrition support (%)	HHs reported improvement in HH Nutrition	0	21	84
CI 1.2.8 Women reporting MDDW	Women DDS	Not measured	Improved (score not pooled)	9.07
CI 1.2.9 HHs with improved KAP	Food hygiene behavior index	Not measured	Not measured	2.07
CI 2.2.1 Beneficiaries with new jobs/employment (%)	HHs reporting income increase	0	73	86.6
CI 2.2.2 Supported rural enterprises reporting an increase in profit (%)	VACs reported profit increase	0	Not measured	100
	Avg. profit per PO (PKR/year)	0	Not measured	212,368
CI 2.1.3 Rural producers' organizations/VACs supported	No. of POs/VACs	0	162	162
CI 2.1.4 HHs members of producer orgs/VACs (Numbers)	Members	0	40,613	40,613
CI 2.2.3 POs with contracts/buyer linkages (%)	POs	0	0	33.3
CI 2.1.5 Roads constructed/rehabilitated (Km)	Length of FMRs	0	400	400
CI 2.2.6 Households reporting improved physical access to markets (hours)	Avg. travel time saved	0	Not measured	3.5

Table Annex B: Main uses of additional income

Intervention Category	Nutrition Food	Farm/Business Investment	Education Expenditure	Purchase of Assets	Purchase of Animals	Home Improvement	Medical/Health	Social Occasions	Savings	Other
% HH	31.15	23.46	64.80	5.03	8.10	37.43	48.04	27.23	8.94	7.40

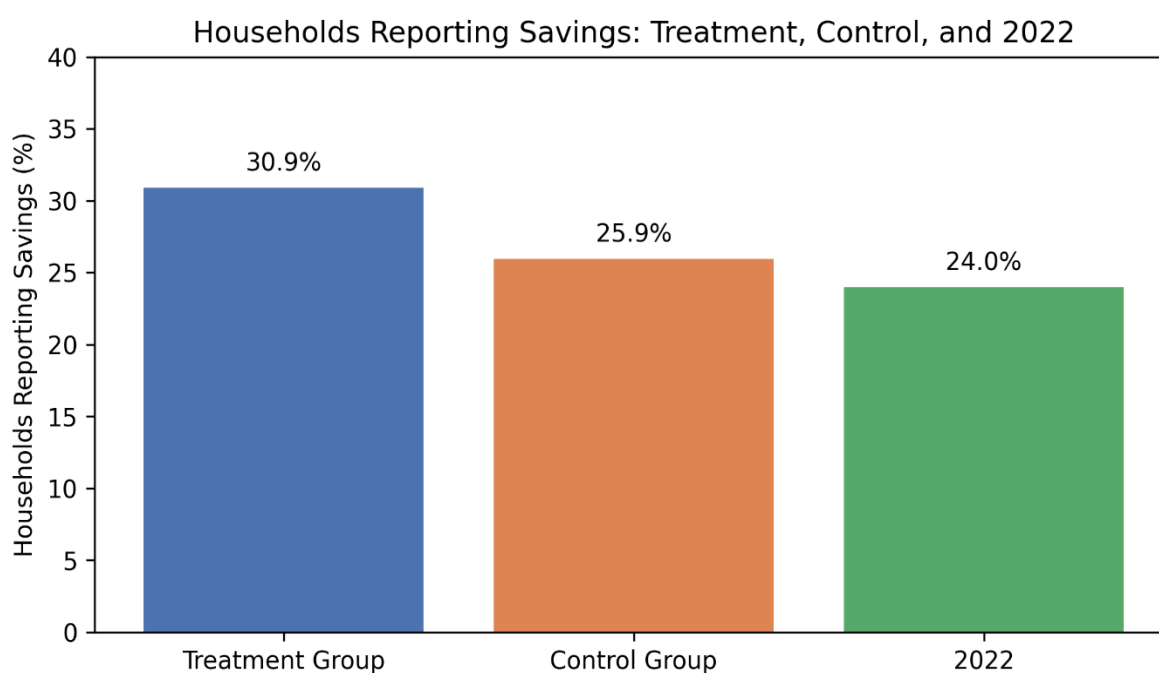


Figure Annex-A

Table Annex-C: Savings trend and uses between treatment and control groups

Savings Indicator (Yes %)	Treatment Group (%)	Control Group (%)
Saved in past 12 months	31.00	25.95
Invest savings in income-generating activities	26.80	17.65
Use savings to purchase tools/raw materials/inputs	22.80	19.12
Use savings for home improvement	61.20	64.22
Savings for children's education	73.60	80.39
Savings for emergencies/health	68.80	74.02
Savings for social occasions	34.40	30.88
Other purposes	7.20	7.84

Table Annex-D: Households' Assessment of the Project's Role in Enabling Savings

Intervention Category	Major Contribution (%)	Large Contribution (%)	Some Contribution (%)	Small Contribution (%)
	2022 → 2025	2022 → 2025	2022 → 2025	2022 → 2025
ILD only	18.00 → 21.88	32.00 → 15.63	41.00 → 50.00	9.00 → 12.50
FMR only	38.00 → 18.37	21.00 → 40.82	36.00 → 36.73	4.00 → 4.08
VC only	16.00 → 30.10	27.00 → 24.27	45.00 → 42.72	11.00 → 2.91
ILD + FMR	4.00 → 20.00	33.00 → 13.33	44.00 → 60.00	19.00 → 6.67
ILD + VC	22.00 → 0.00	30.00 → 0.00	48.00 → 100.00	0.00 → 0.00
ILD + FMR + VC	21.00 → 28.57	42.00 → 57.14	26.00 → 14.29	11.00 → 0.00
Overall	21.00 → 25.11	29.00 → 26.03	41.00 → 44.29	1.00 → 4.57

